



YFG HL THERMAL GLUE DISPENSING MACHINE

Manufactured by:

Paveway Automation Pvt. Ltd.
Sanghvi Steel Complex,
128/2, TELCO Road,
Chinchwad,
Pune - 411019

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YFG HL THERMAL GLUE DISPENSING MACHINE

MARELLI MOTHERSON AUTO SYSTEM LTD.
GAT NR. 148-150, VILLAGE AMBETHAN TALUKA KHED,
NEAR DWARKA SCHOOL, PUNE, MAHARASHTRA- 410501



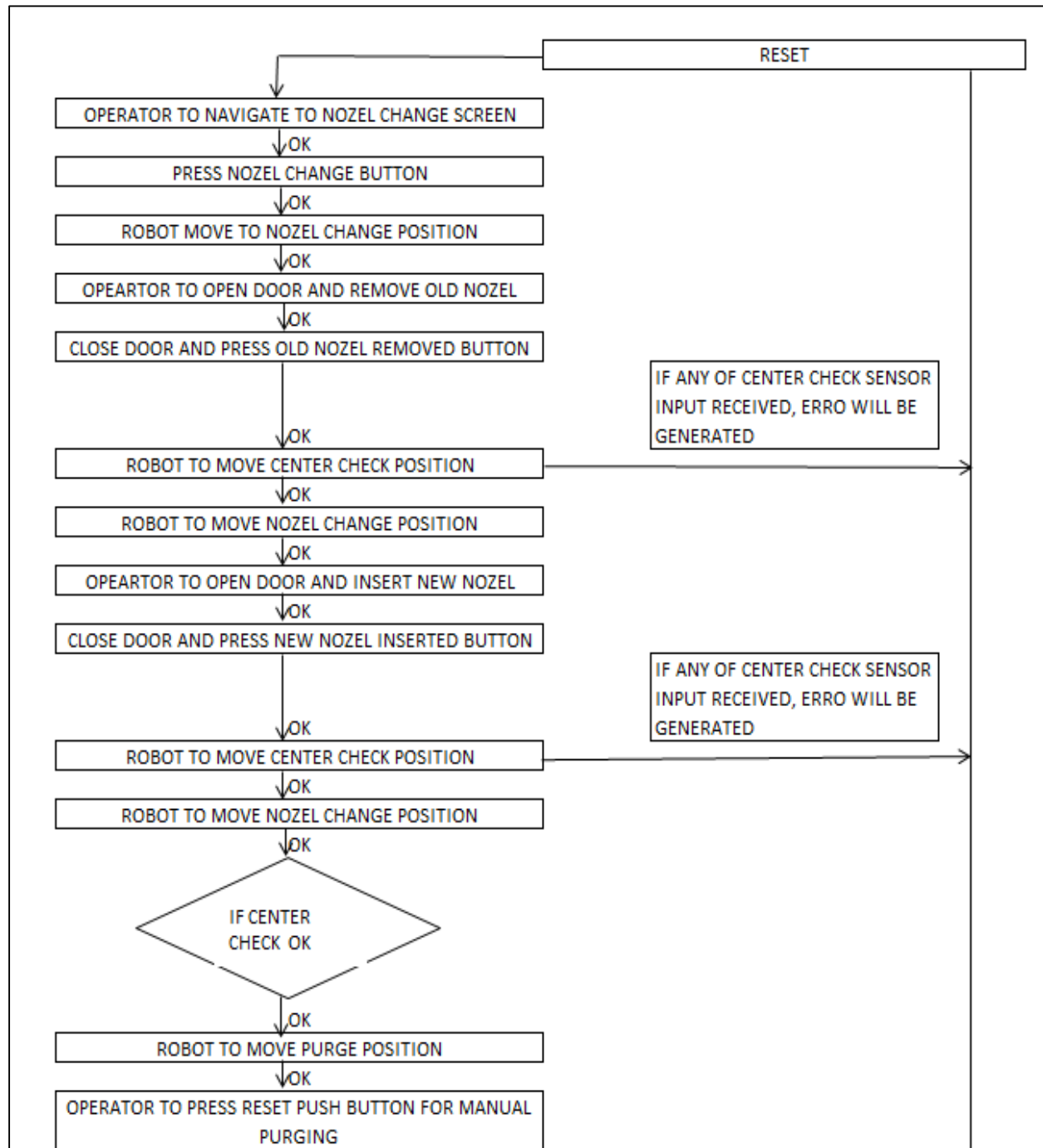
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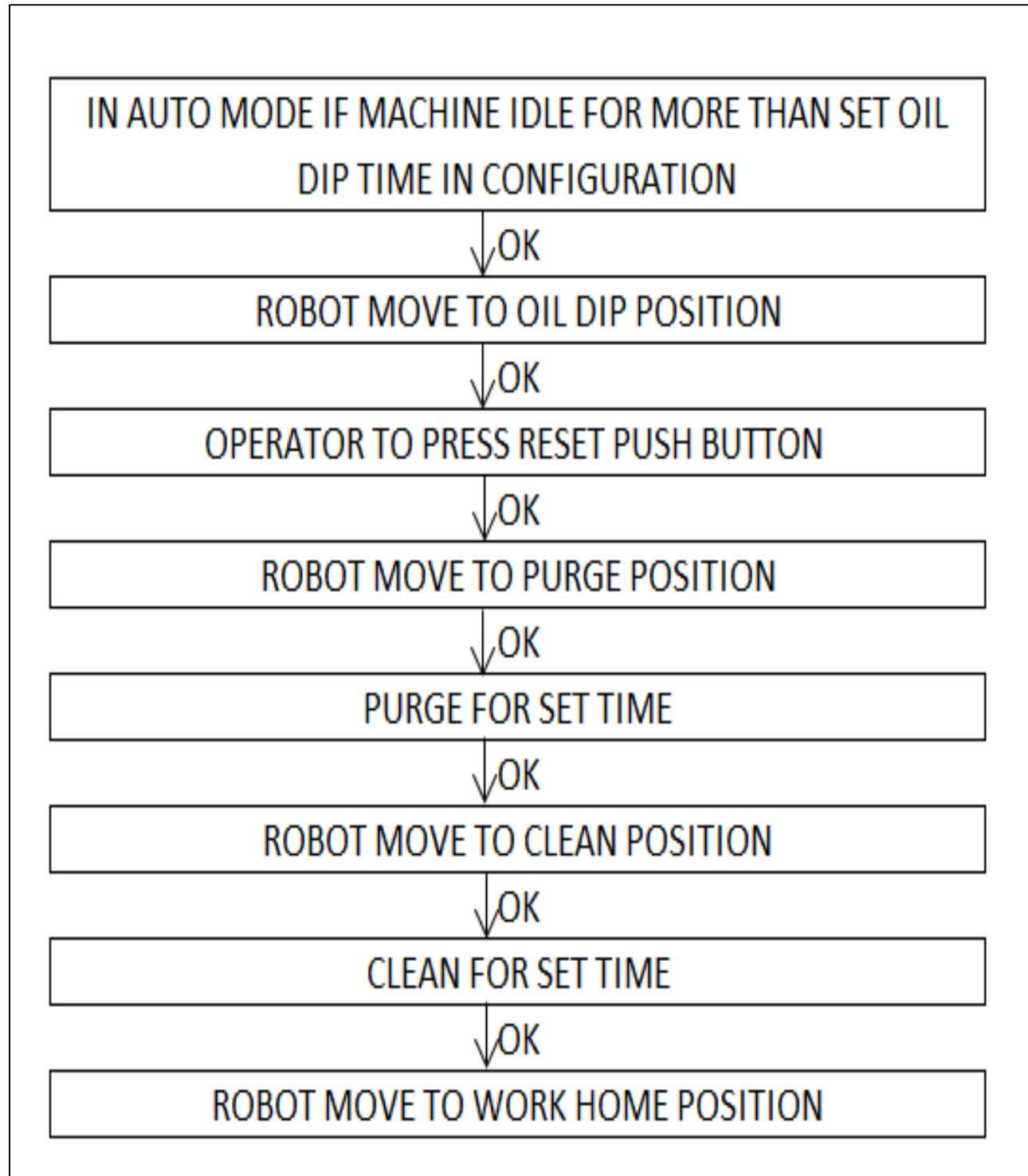
SYSTEM INFORMATION:

Name :	DUV HL THERMAL GLUE DISPENSING MACHINE.
Customer :	MARELLI MOTHERSON AUTO SYSTEM LTD. GAT NR. 148-150, VILLAGE AMBETHAN TALUKA KHED, NEAR DWARKA SCHOOL, PUNE, MAHARASHTRA- 410501
Manufacturing Date :	August,2021
Supply Voltage :	415V AC
Power consumption:	2.6KW
Manufacturer :	Paveway Automation Pvt. Ltd. Sanghvi Steel Complex, 128/2, Telco Road, Chinchwad, Pune- 411019. tel: 020-27450840
Email ID :	applications@pavewayautomation.com

PROCESS SEQUENCE:

NOZZLE CHANGE AND WEIGHT CHECK CYCLE:





IN AUTO MODE IF MACHINE IDLE FOR MORE THAN SET PURGE
TIME IN CONFIGURATION

↓ OK

ROBOT MOVE TO PURGE POSITION

↓ OK

PURGE FOR SET TIME

↓ OK

ROBOT MOVE TO CLEAN POSITION

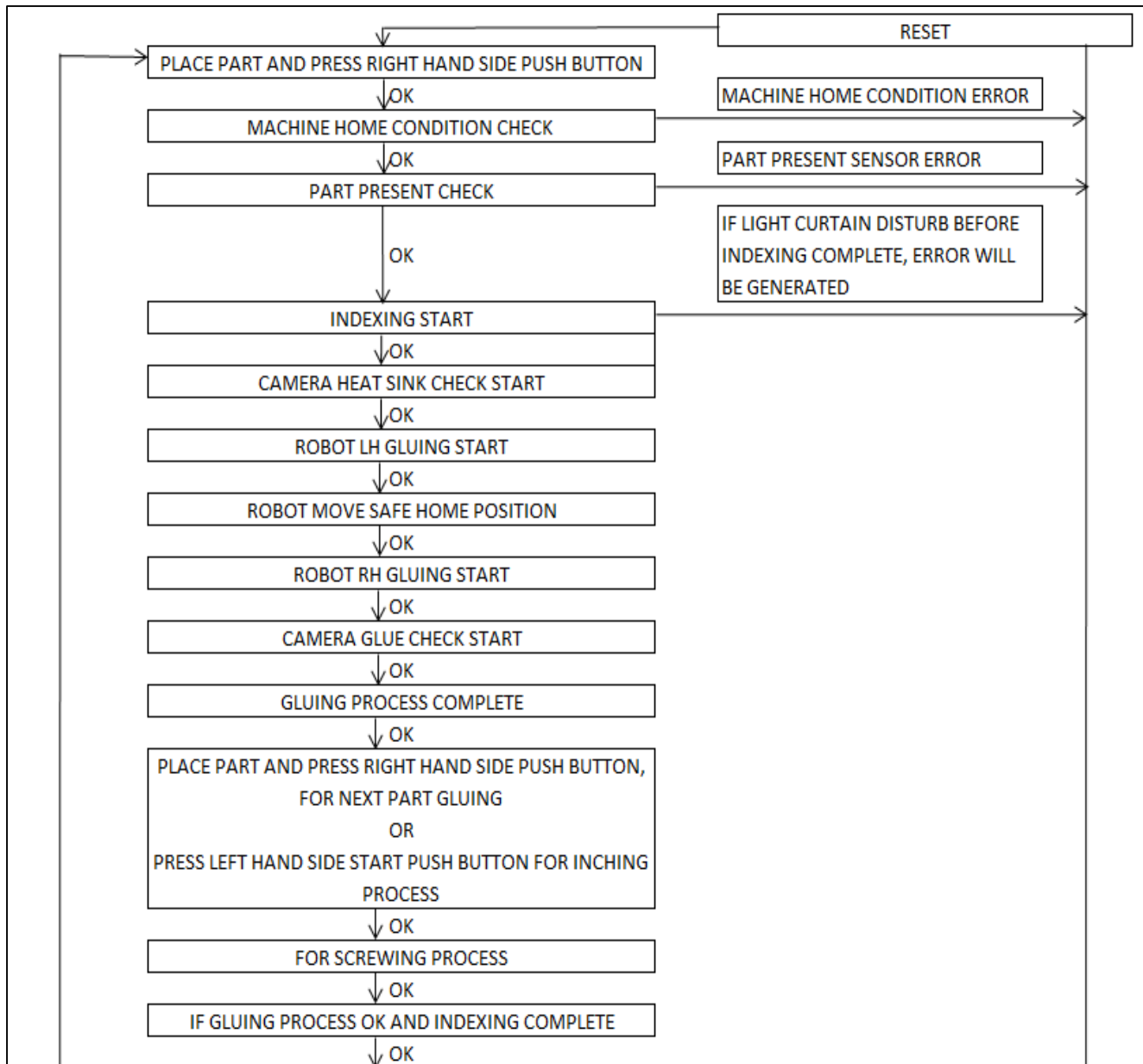
↓ OK

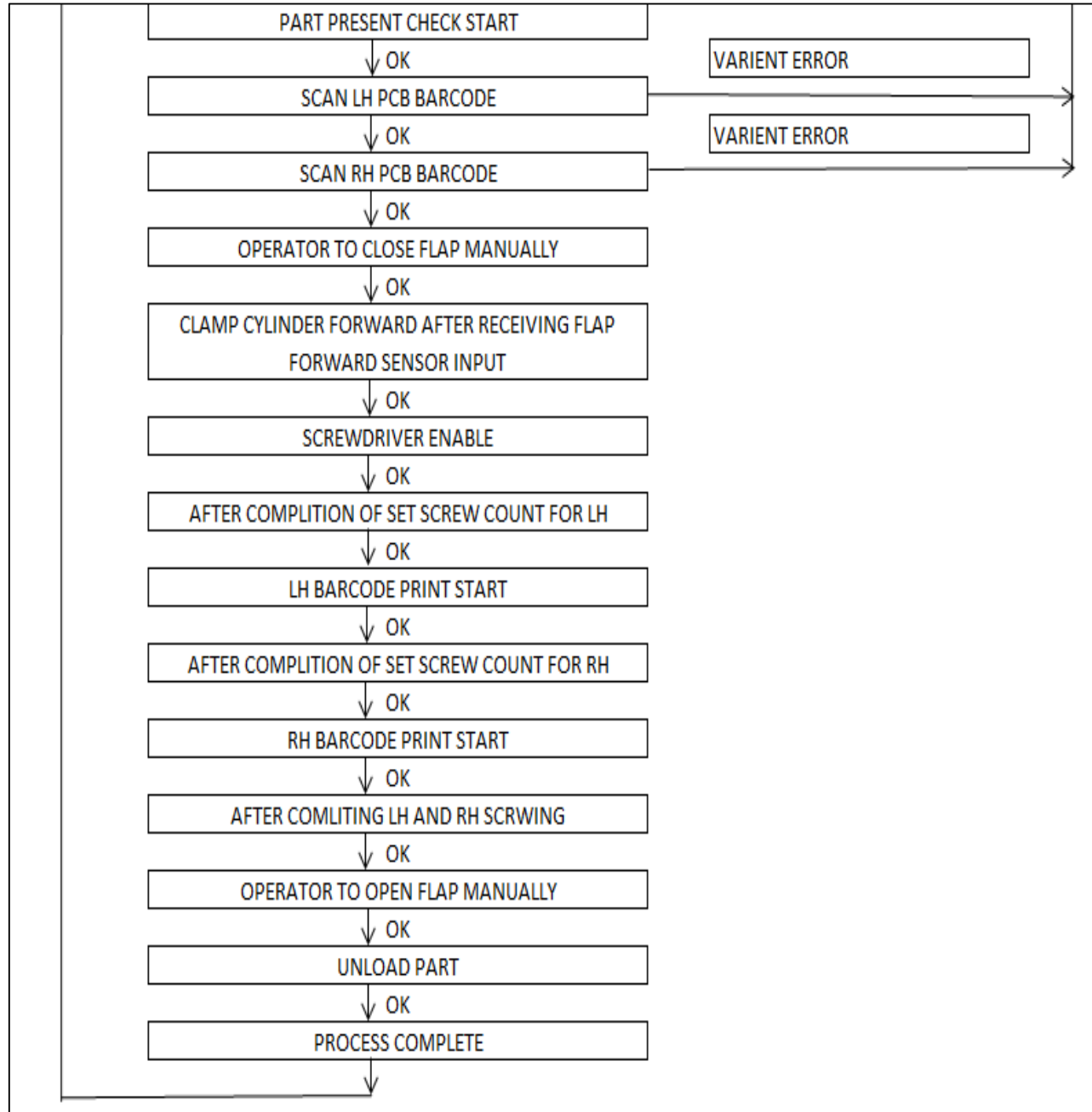
CLEAN FOR SET TIME

↓ OK

ROBOT MOVE TO WORK HOME POSITION

AUTO CYCLE PROCESS:

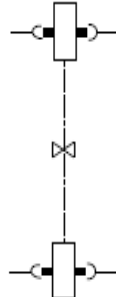
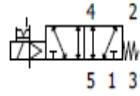
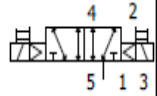




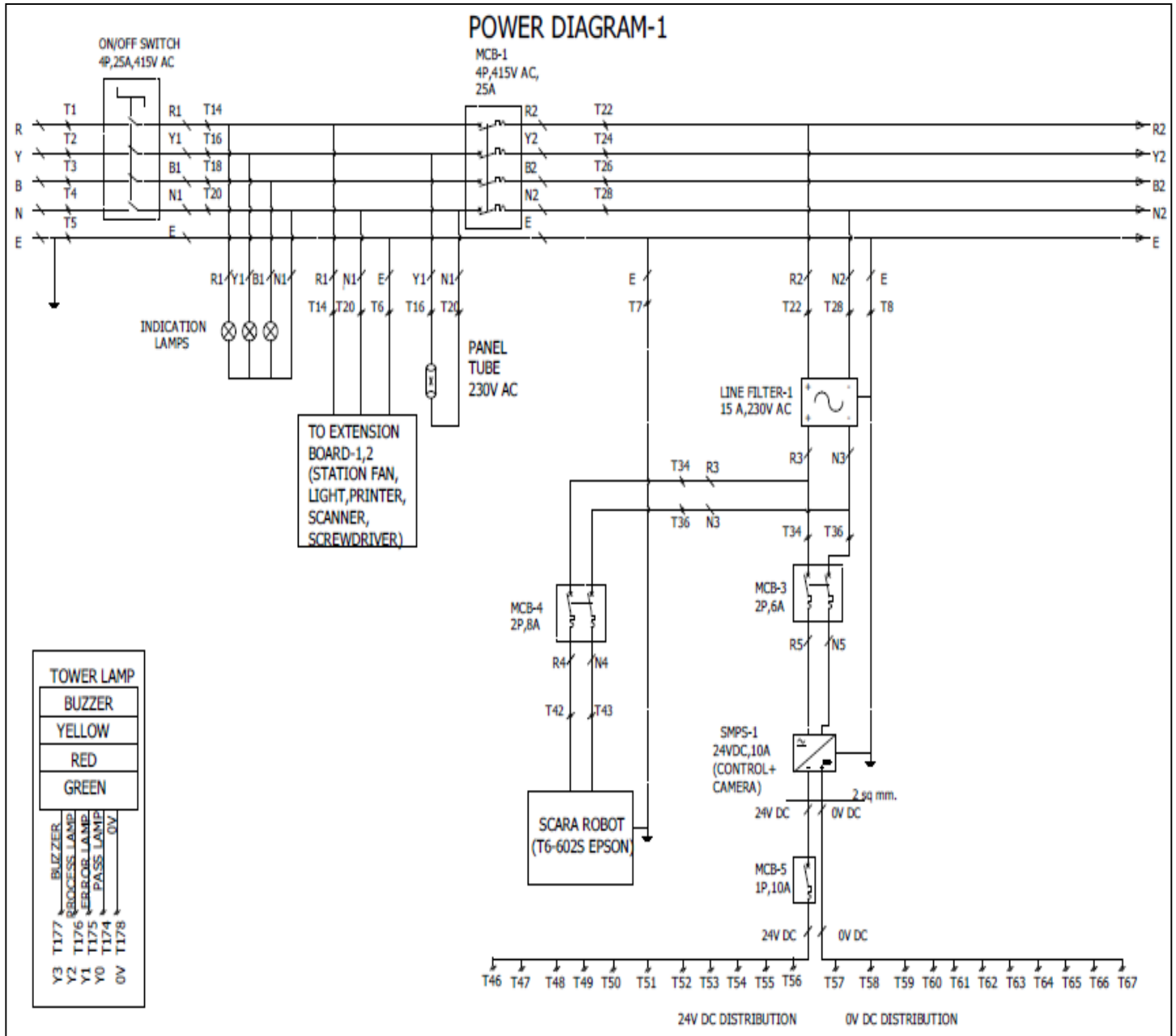
WIRING DIAGRAMS:

SYMBOL DESCRIPTION

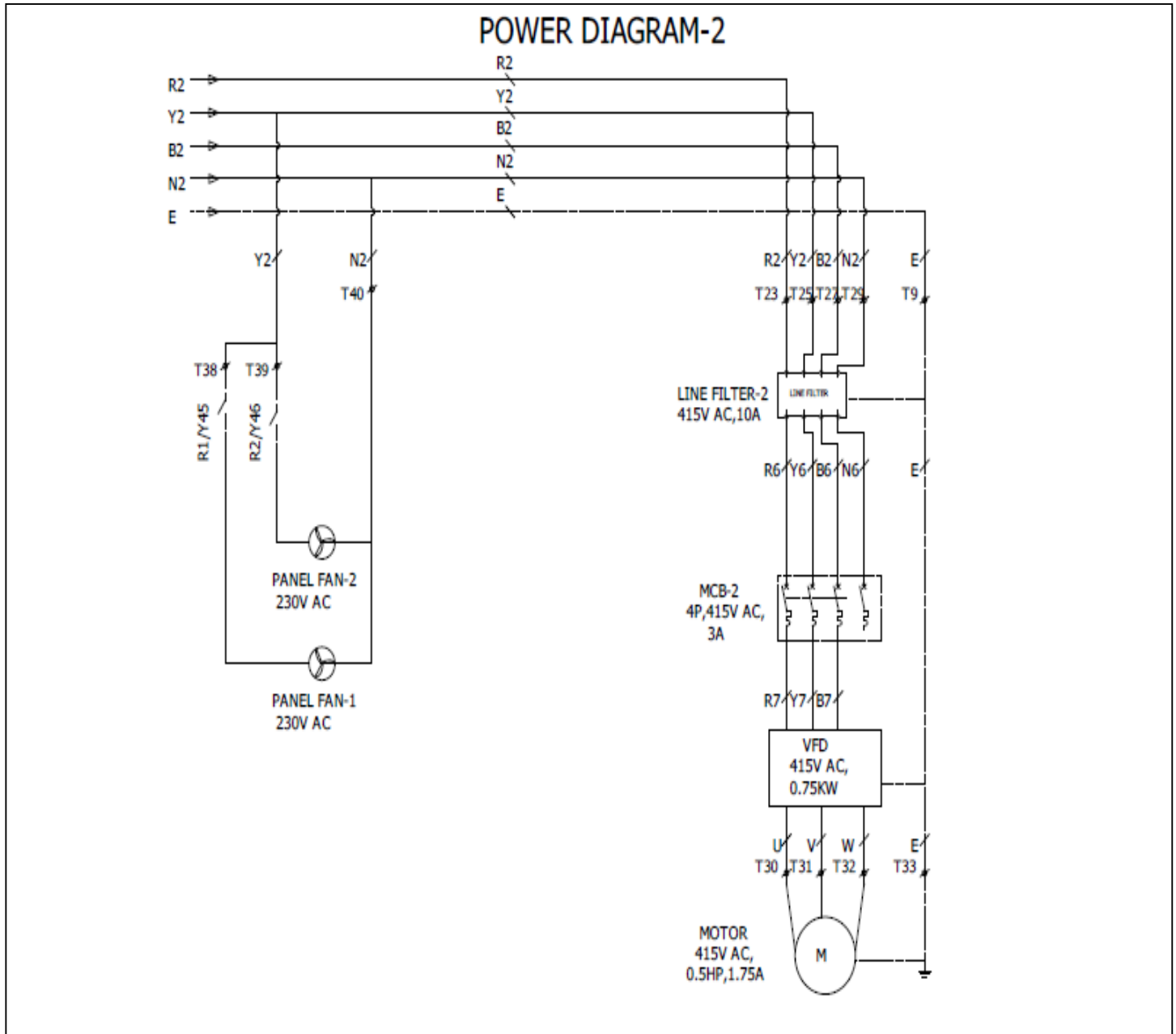
SYMBOL DESCRIPTION

Symbol	Symbol description	Page nos.	Page description
	Earthing		Airoff
	2 Pole MCB		
	2 Pole on/off switch		Double acting solenoid valve
	Lamp		
	Tubelight		
	Fan		
	SMPS		
	1 pole MCB		Buzzer
	Fuse		
	Push button		single acting solenoid valve 5/2 way- pneumatic
	Emergency stop button		
	Proximity sensor		Double acting solenoid valve 5/2 way- pneumatic
	Limit switch		
	Relay coil		

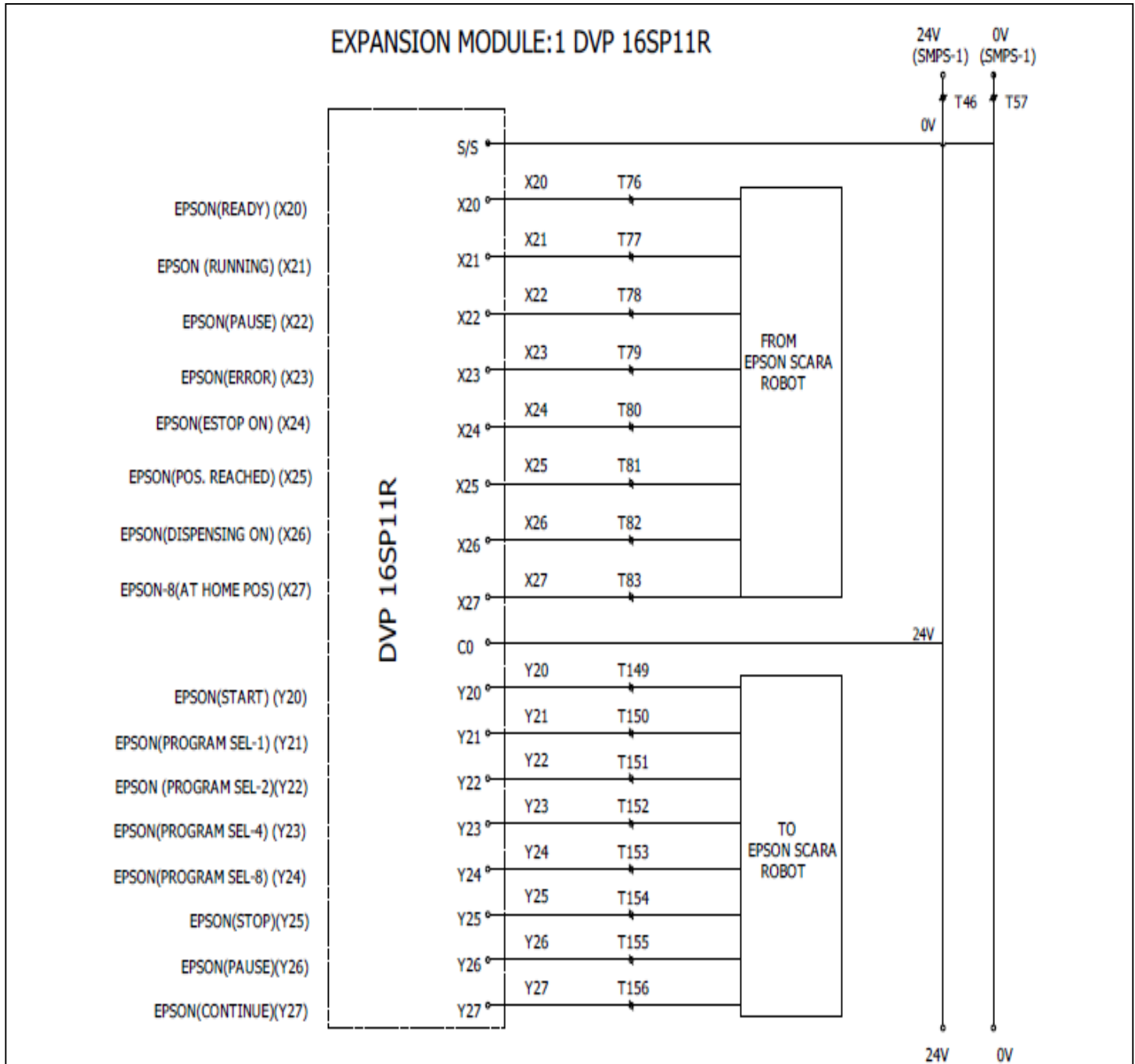
POWER DIAGRAM-1



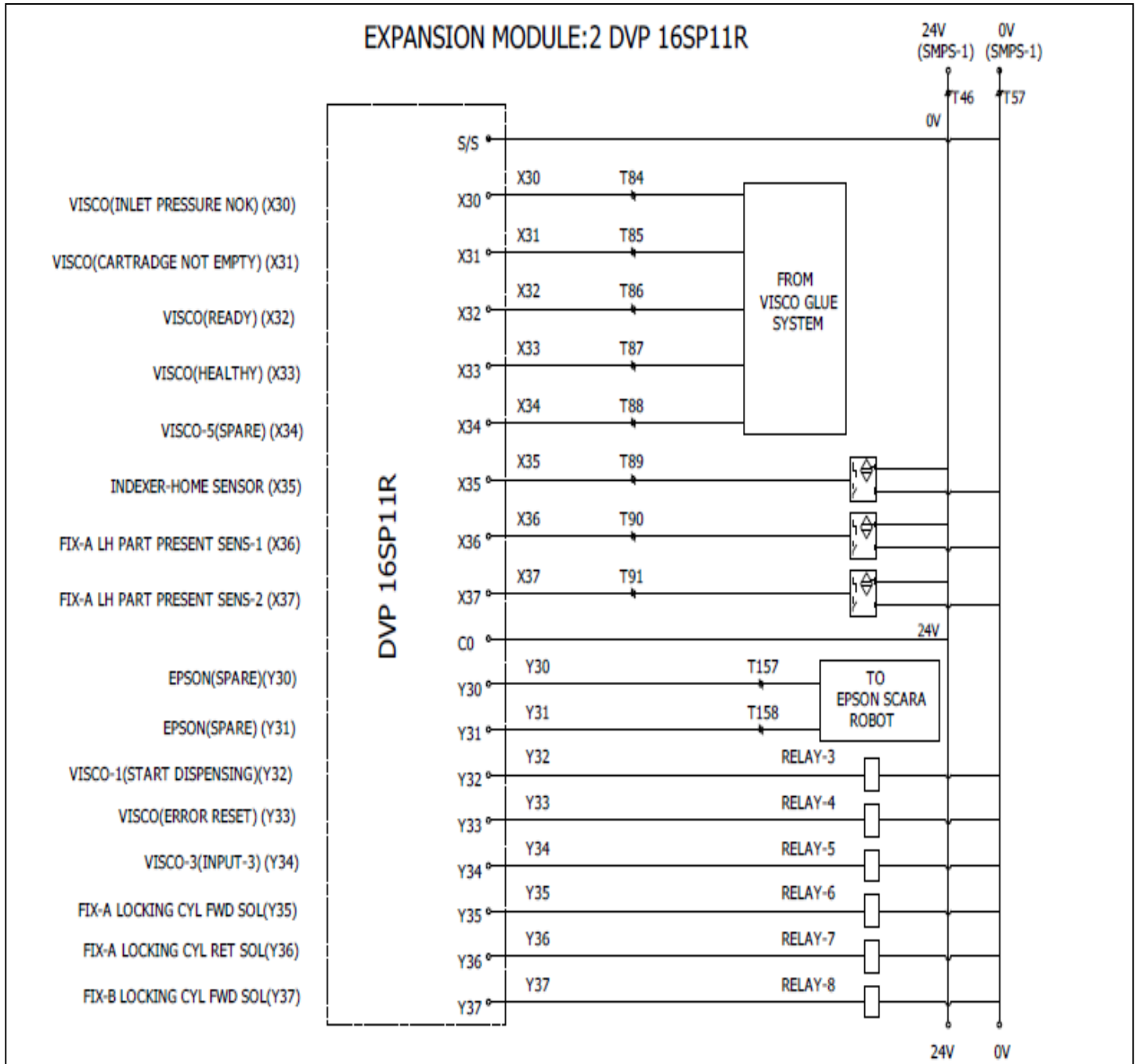
POWER DIAGRAM-2



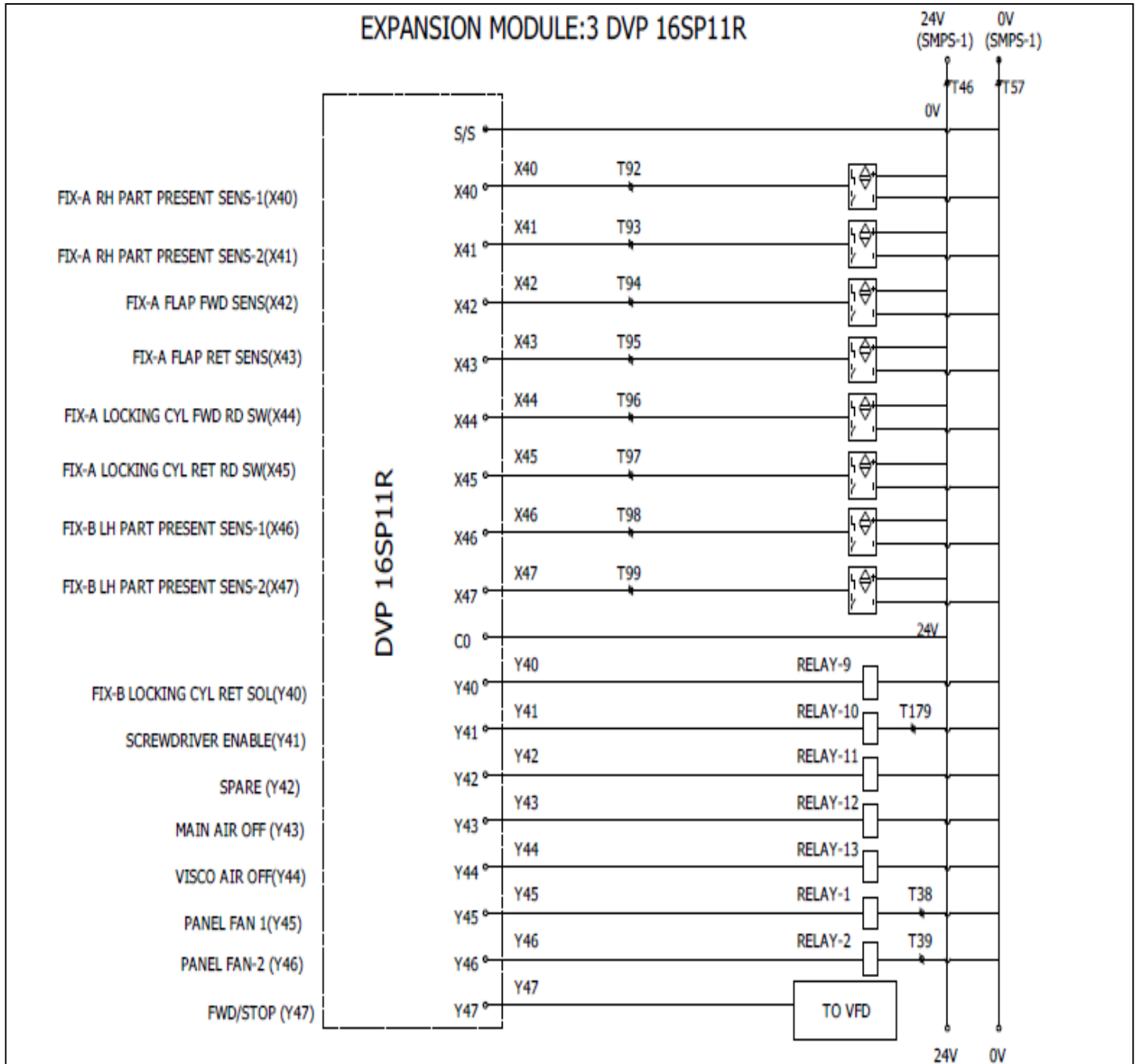
EXPANSION MODULE 1: DVP16SP11R



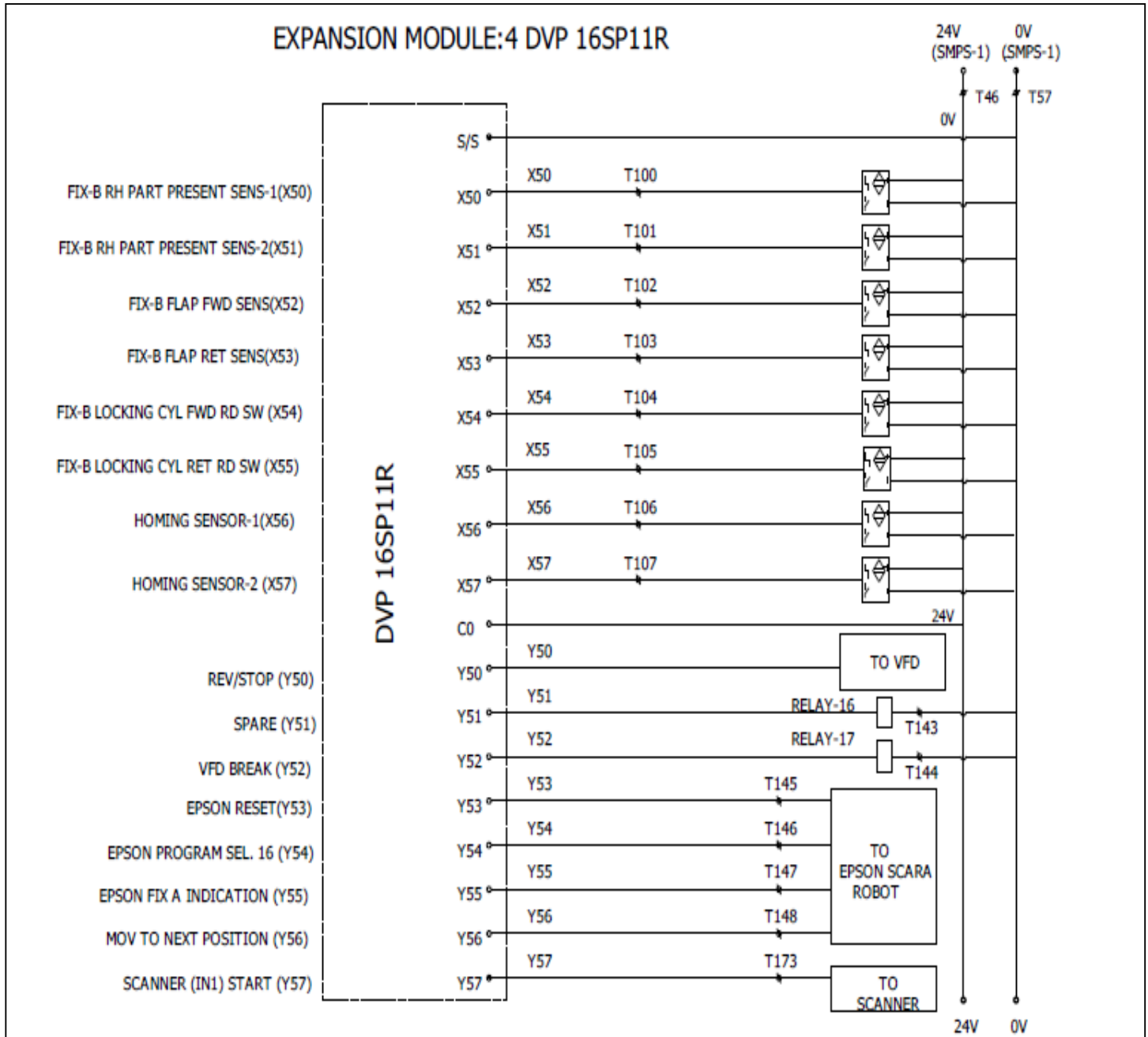
EXPANSION MODULE 2: DVP16SP11R



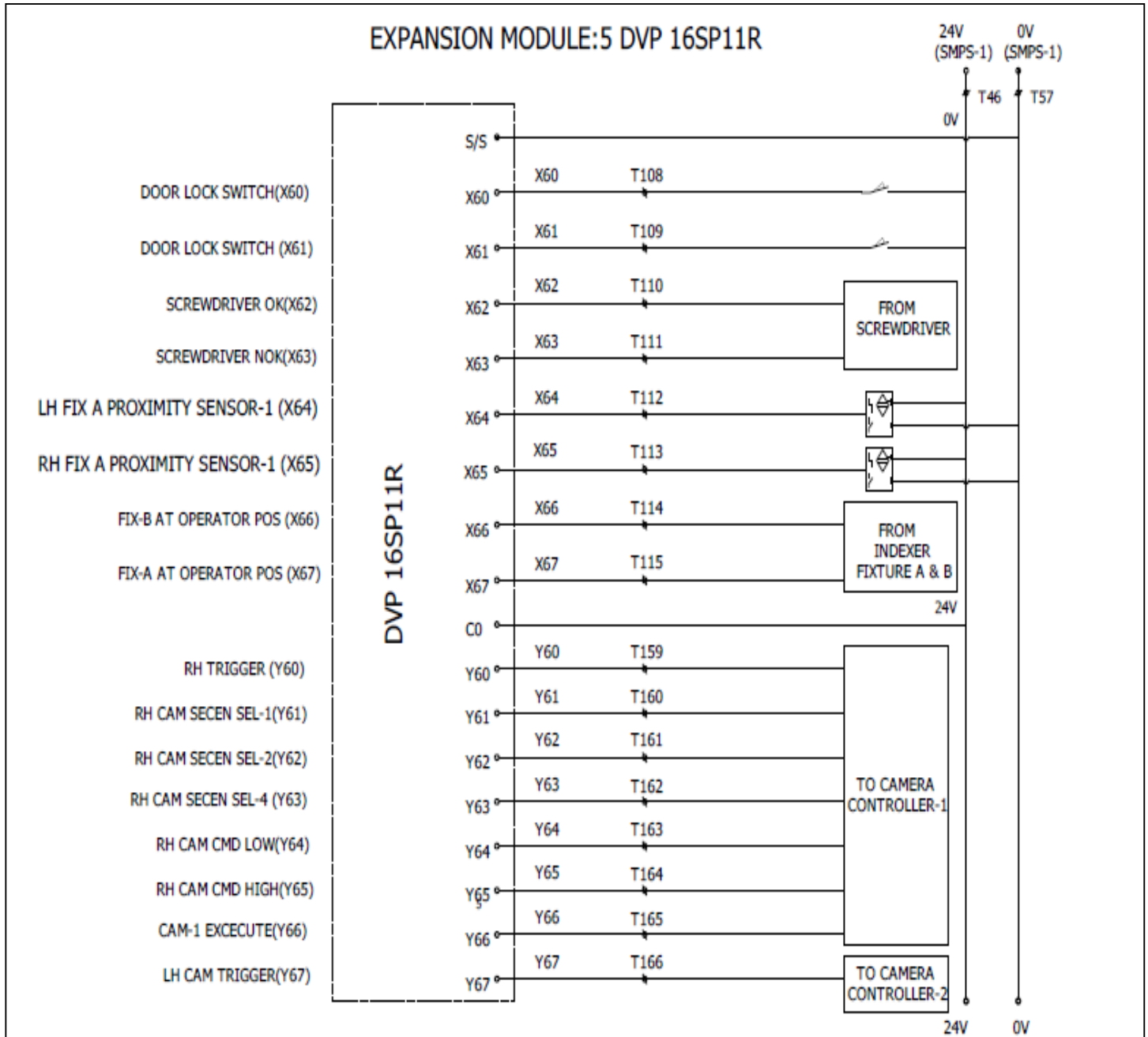
EXPANSION MODULE-3 DVP 16SP11R



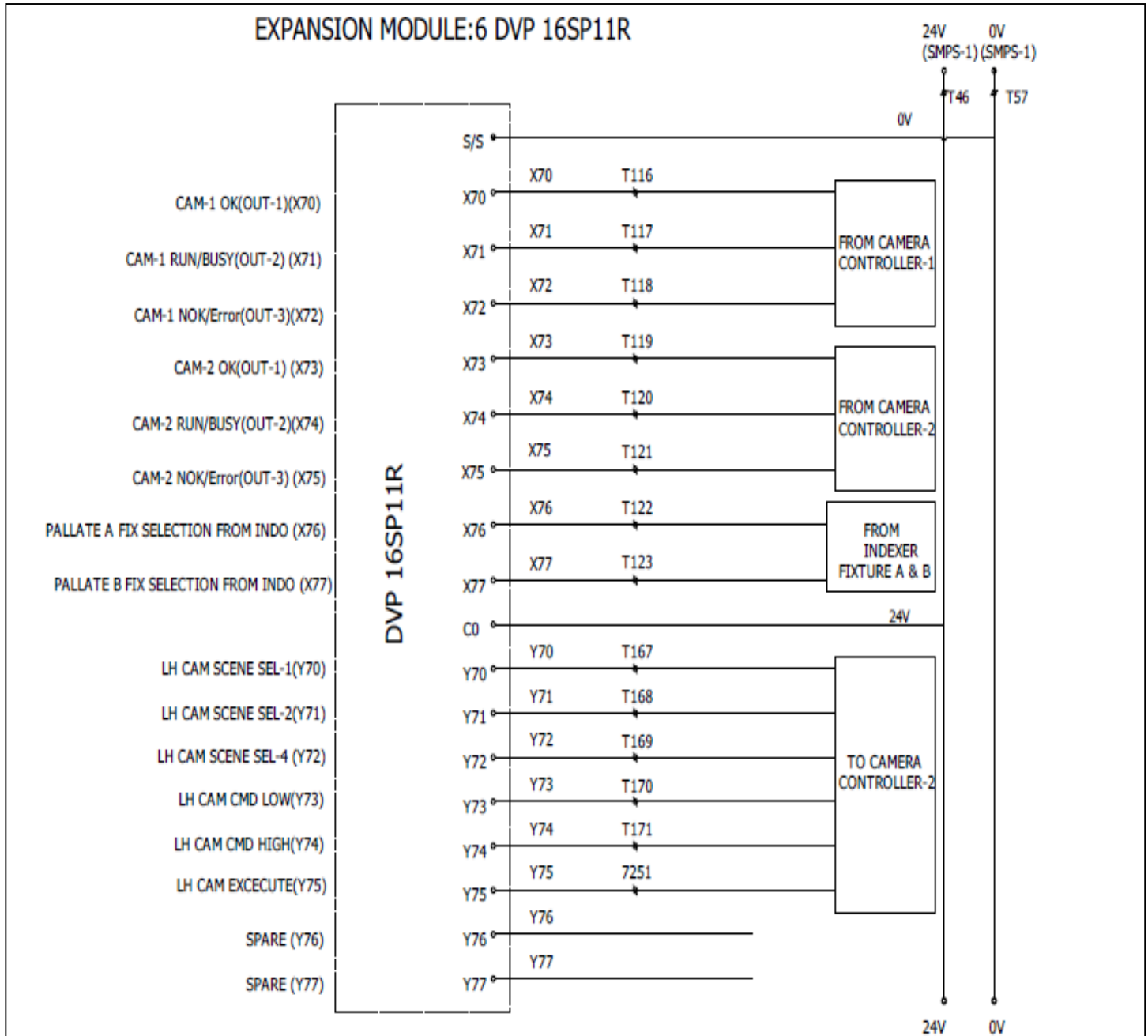
EXPANSION MODULE-4 DVP 16SP11R



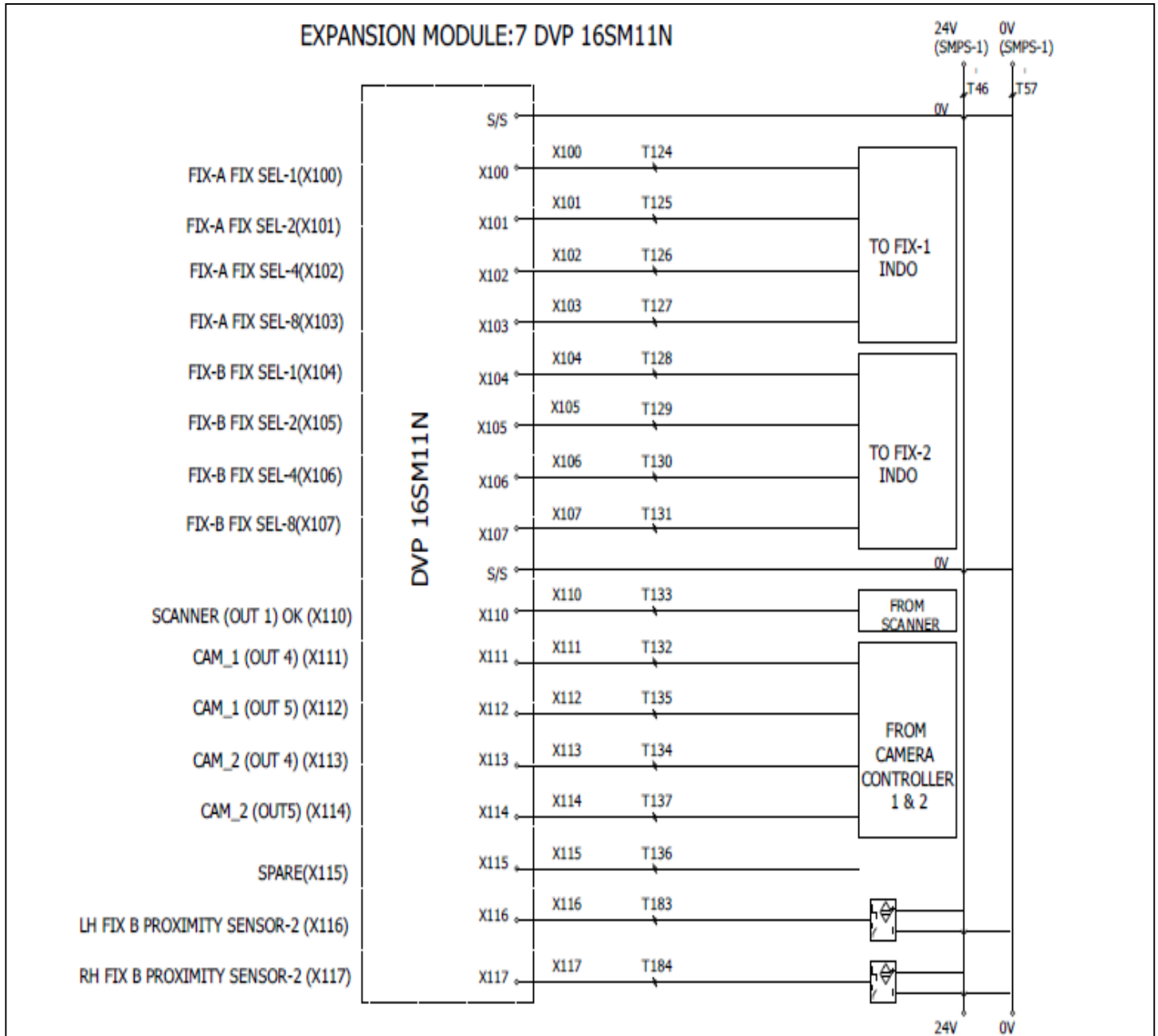
EXPANSION MODULE-5 DVP16SP11R



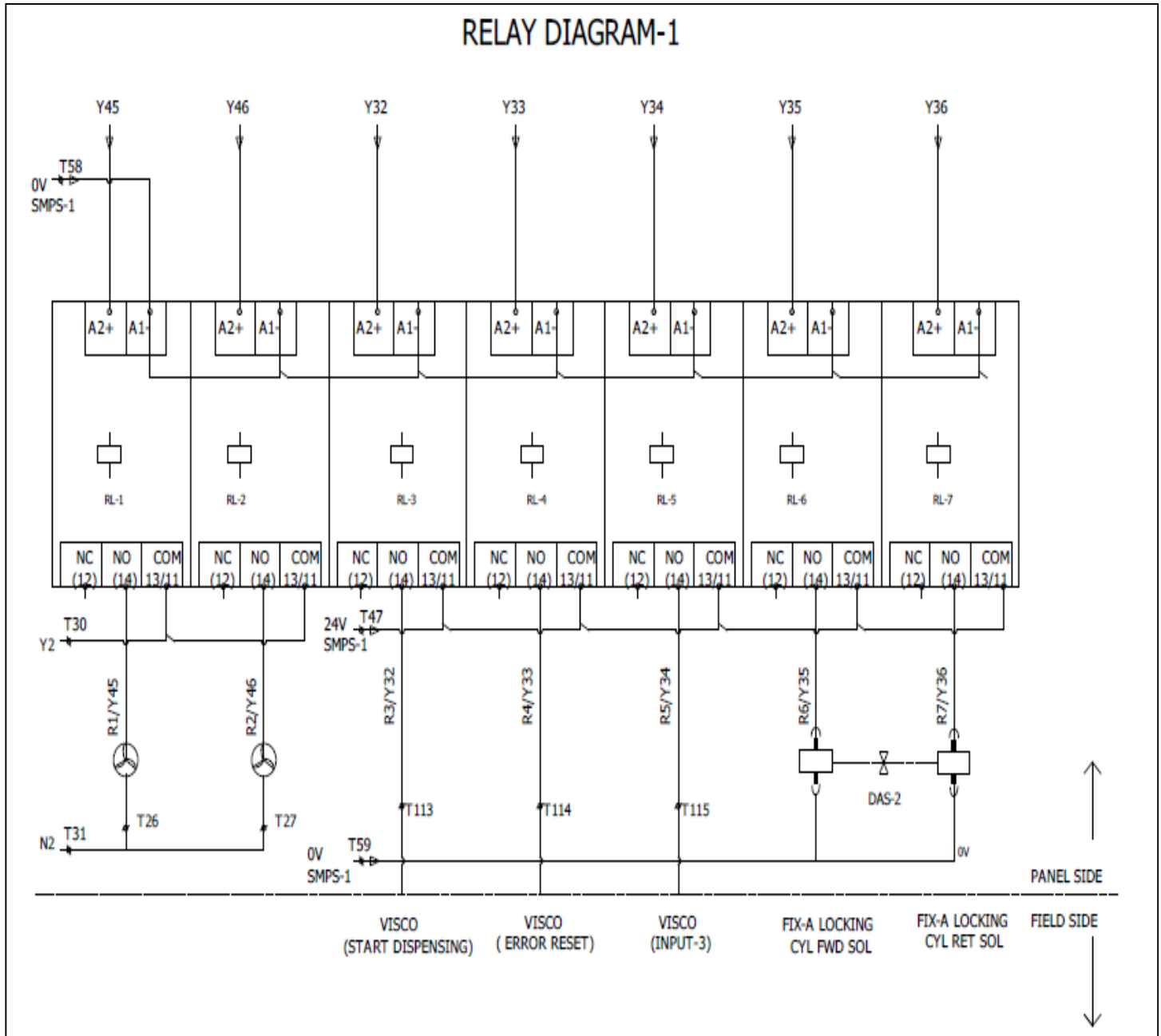
EXPANSION MODULE-6 DVP16SP11R



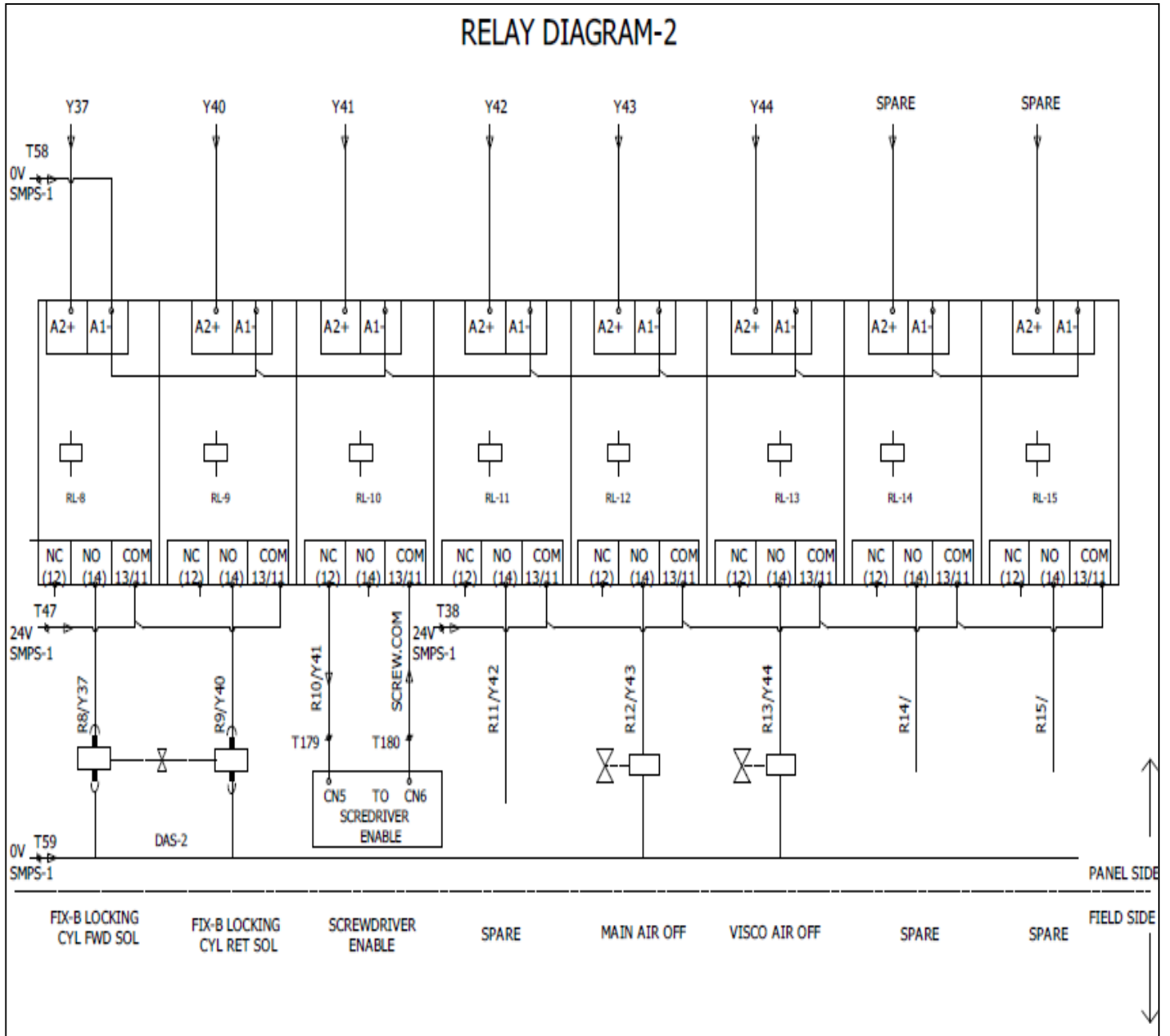
EXPANSION MODULE-7 DVP16SM11N



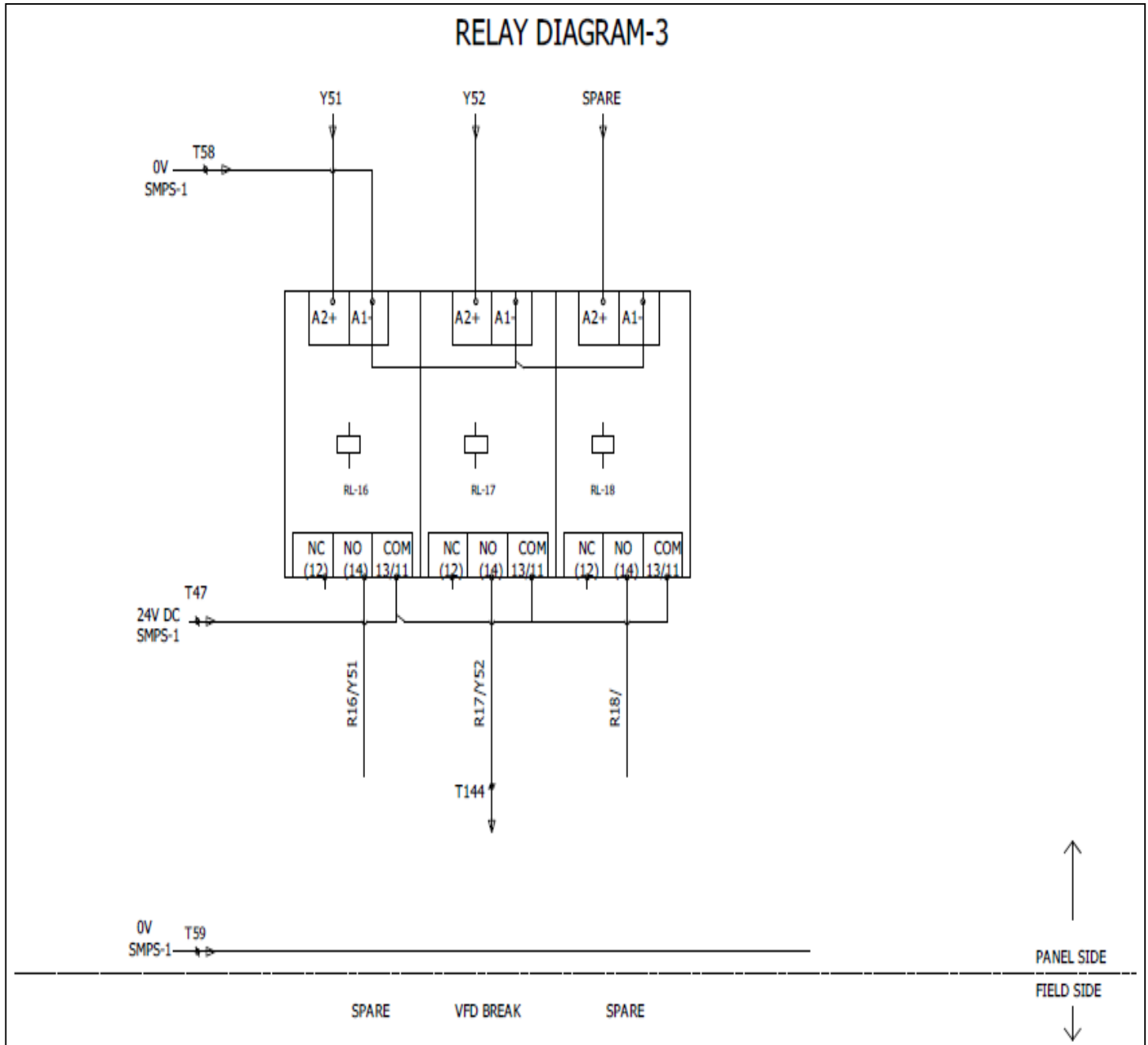
RELAY DIAGRAM-1



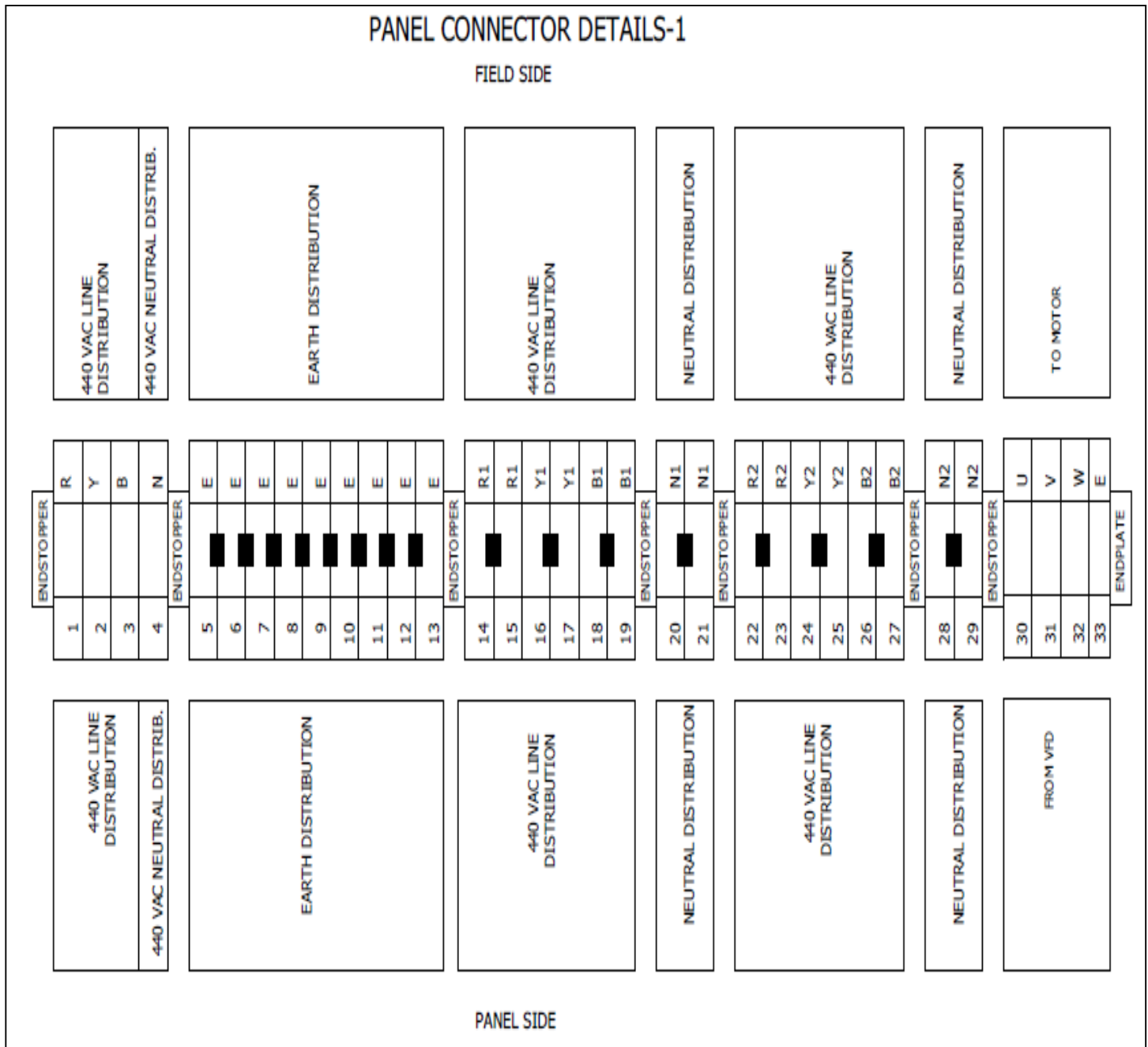
RELAY DIAGRAM-2



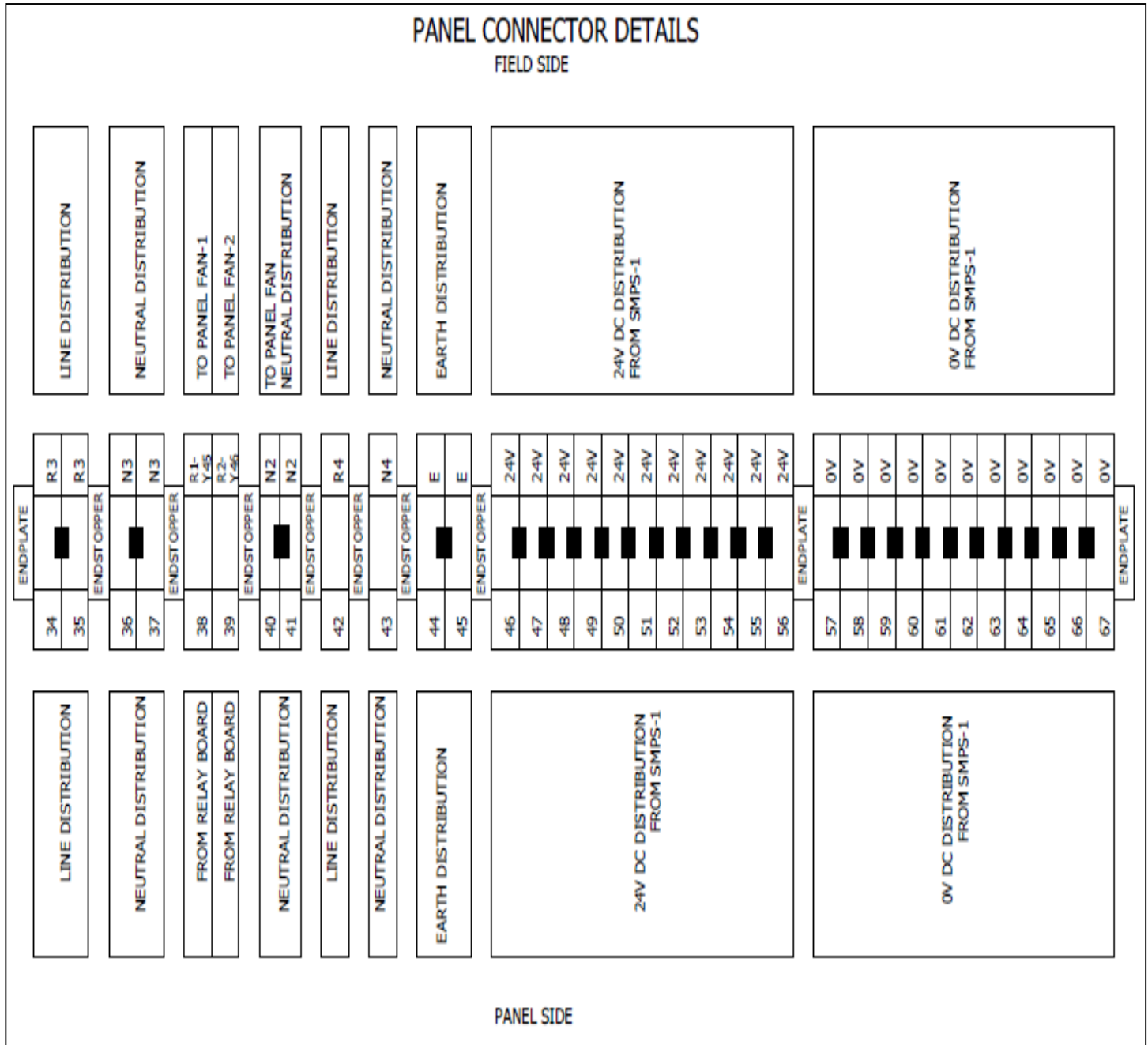
RELAY DIAGRAM-3



PANEL CONNECTOR DETAILS-1



PANEL CONNECTOR DETAILS-2



PANEL CONNECTOR DETAILS-3

PANEL CONNECTOR DETAILS											
FIELD SIDE											
PANEL SIDE	START PB 2(X1)	X1	69	START PB 1(X0)	X1	69	EMERGENCY STOP PB (X2)				
	RESET PB (X3)	X3	71	MAIN DIGITAL PRESSURE SW (X4)	X3	71					
	VISCO DIGITAL PRESSURE SW (X5)	X5	73	LIGHT CURTAIN-1 (X6)	X5	73					
	REJECTION BIN SENSOR R (X7)	X7	75	EPSON (READY) (X20)	X7	75					
	EPSON (RUNNING) (X21)	X21	77	EPSON (PAUSE) (X22)	X21	77					
	EPSON (ERROR) (X23)	X23	79	EPSON (STOP ON) (X24)	X23	79					
	EPSON (POS REACHED) (X25)	X25	81	EPSON (DISPENSING ON) (X26)	X25	81					
	EPSON-8 (AT HOME POSITION) (X27)	X27	83	VISCO (INFLATE PRESSURE NOK) (X30)	X26	83					
	VISCO (CARTRIDGE NOT EMPTY) (X31)	X31	85	VISCO (READY) (X32)	X31	85					
	VISCO (HEALTHY) (X33)	X33	87	VISCO-5 (SPARE) (X34)	X33	87					
	INDEXER- HOME SENSOR (X35)	X35	89	FIXTURE-A LH PART PRESENT SENSOR-1 (X36)	X35	89					
	FIXTURE-A LH PART PRESENT SENSOR-2 (X37)	X37	91	FIXTURE-A RH PART PRESENT SENSOR-1 (X40)	X37	91					
	FIXTURE-A RH PART PRESENT SENSOR-2 (X41)	X41	93	FIXTURE A FLAP FWD SENSOR (X42)	X41	93					
	FIXTURE A FLAP RETURN SENSOR (X43)	X43	95	FIXTURE A LOCKING CYL FWD RD SW (X44)	X43	95					
	FIXTURE A LOCKING CYL RET RD SW (X45)	X45	97	FIXTURE B LH PART PRESENT SENSOR-1 (X46)	X45	97					
	FIXTURE B LH PART PRESENT SENSOR-2 (X47)	X47	99		X47	99					
	FIXTURE B RH PART PRESENT SENSOR-2 (X51)	X51	101	FIXTURE B RH PART PRESENT SENSOR-1 (X50)							
	FIXTURE B FLAP RET SENSOR (X53)	X53	103	FIXTURE B FLAP FWD SENSOR (X52)	X51	101					
	FIXTURE B LOCKING CYL RET RD SW (X55)	X55	105	FIXTURE B LOCKING CYL FWD RD SW (X54)	X53	103					
HOMING SENSOR-2 (X57)	X57	107	HOMING SENSOR 1 (X56)	X55	105						
				X57	107						
				ENDPLATE							

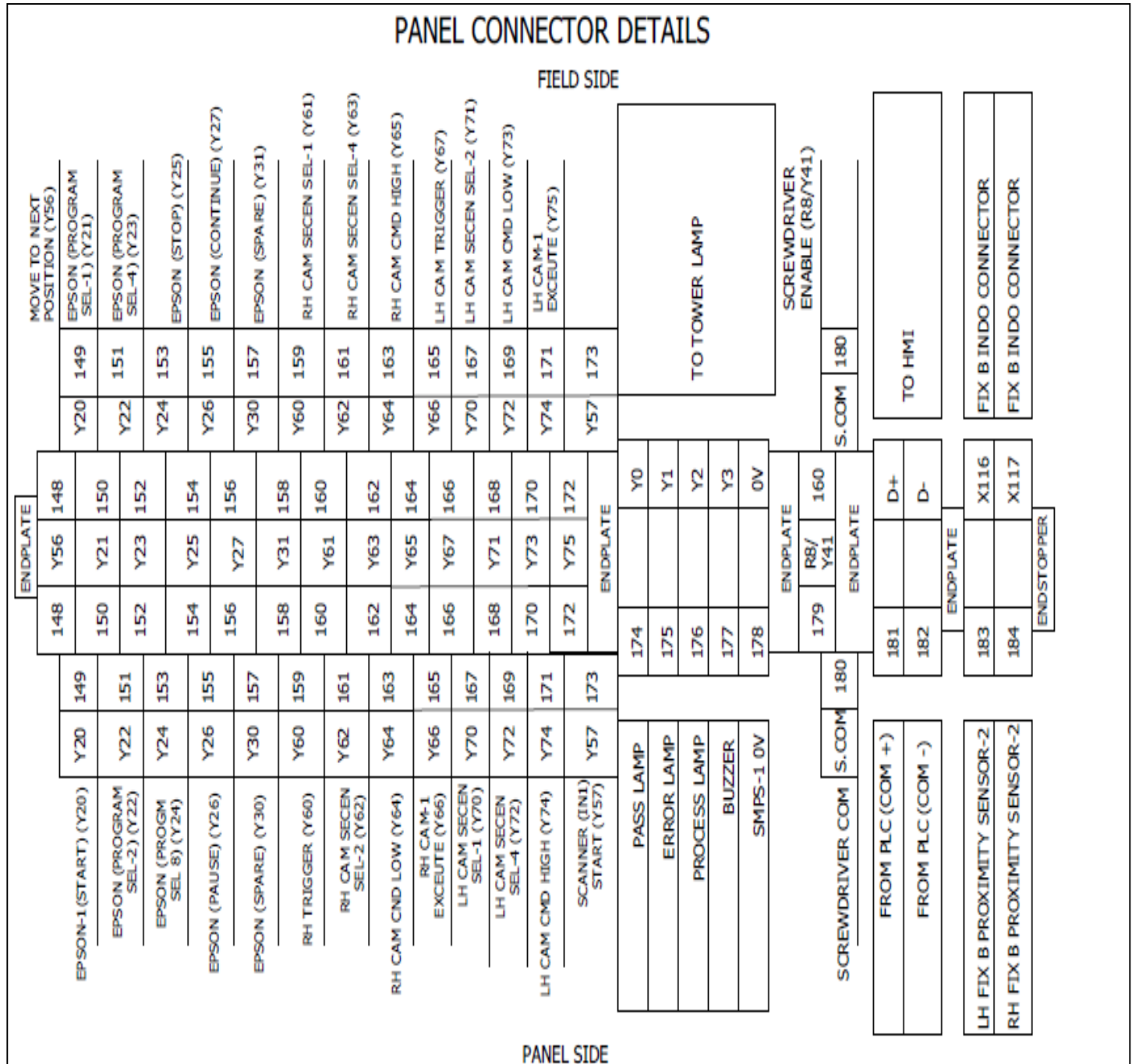
PANEL CONNECTOR DETAILS-4

PANEL CONNECTOR DETAILS

FIELD SIDE

ENDPLATE				DOOR LOCK SW 1 (X60)				SCREWDRIVER OK (X62)				LH FIX A PROXIMITY SENSOR-1 (X64)				FIXTURE-B AT OPERATOR POS (X66)				CAM-1 OK (OUT-1) (X70)				CAM-1 NOK/ERROR (OUT 3) (X72)				CAM-2 RUN/BUSY (OUT 2) (X74)				PALLATE A FIXTURE FROM INDO (X76)				FIXTURE A FIXTURE SELECTION-1 (X100)				FIXTURE A FIXTURE SELECTION-4 (X102)				FIXTURE B FIXTURE SELECTION-1 (X104)				FIXTURE B FIXTURE SELECTION-4 (X106)				CAM-1 (OUT-4) (X111)				CAM-2 (OUT-4) (X113)				SPARE (X115)				VISCO-1 (START DISPENSING) (Y32)				VISCO-3 (INPUT 3) (Y34)				SPARE				VFD BREAK (Y52)				EPSON PROGRAM SEL.16 (Y54)																																																																																																																																																																																											
108	X60	108		X61	109			X63	111			X65	113			X67	115			X71	117			X73	119			X75	121			X77	123			X101	125			X103	127			X105	129			X107	131			X110	133			X112	135			X115	137					Y33	139			Y35	141			Y51	143					Y53	145			Y55	147																																																																																																																																																																																						

PANEL CONNECTOR DETAILS-5



ELECTRICAL JUNCTION BOX

ELECTRICAL JUNCTION BOX

FIELD SIDE

FROM SMPS
FROM SMPS
FIX-A LH PART PRESENT SENS-1
FIX-A LH PART PRESENT SENS-2
FIX-A RH PART PRESENT SENS-1
FIX-A RH PART PRESENT SENS-2
FIX-A FLAP FWD SENS
FIX-A FLAP RET SENS
FIX-A LOCKING CYL FWD RD SW
FIX-A LOCKING CYL RET RD SW
FIX-A LH FIX SEL-1
FIX-A LH FIX SEL-2
FIX-A LH FIX SEL-3
FIX-A LH FIX SEL-4
PALLATE B FIX FROM INDO
FROM SMPS
FROM SMPS
FIX-B LH PART PRESENT SENS-1
FIX-B LH PART PRESENT SENS-2
FIX-B RH PART PRESENT SENS-1
FIX-B RH PART PRESENT SENS-2
FIX-B RH FIX SEL-3
FIX-B RH FIX SEL-4
FIX-B FLAP FWD SENS
FIX-B FLAP RET SENS
FIX-B LOCKING CYL FWD RD SW
FIX-B LOCKING CYL RET RD SW
PALLATE A FIX FROM INDO
FIX-B RH FIX SEL-1
FIX-B RH FIX SEL-2

ENDPLATE

24VDC	1	24VDC
0VDC	2	0VDC
X36	3	X36
X37	4	X37
X40	5	X40
X41	6	X41
X42	7	X42
X43	8	X43
X44	9	X44
X45	10	X45
X100	11	X100
X101	12	X101
X102	13	X102
X103	14	X103
X77	15	X77
24VDC	16	24VDC
0VDC	17	0VDC
X46	18	X46
X47	19	X47
X50	20	X50
X51	21	X51
X106	22	X106
X107	23	X107
X52	24	X52
X53	25	X53
X54	26	X54
X55	27	X55
X76	28	X76
X104	29	X104
X105	30	X105

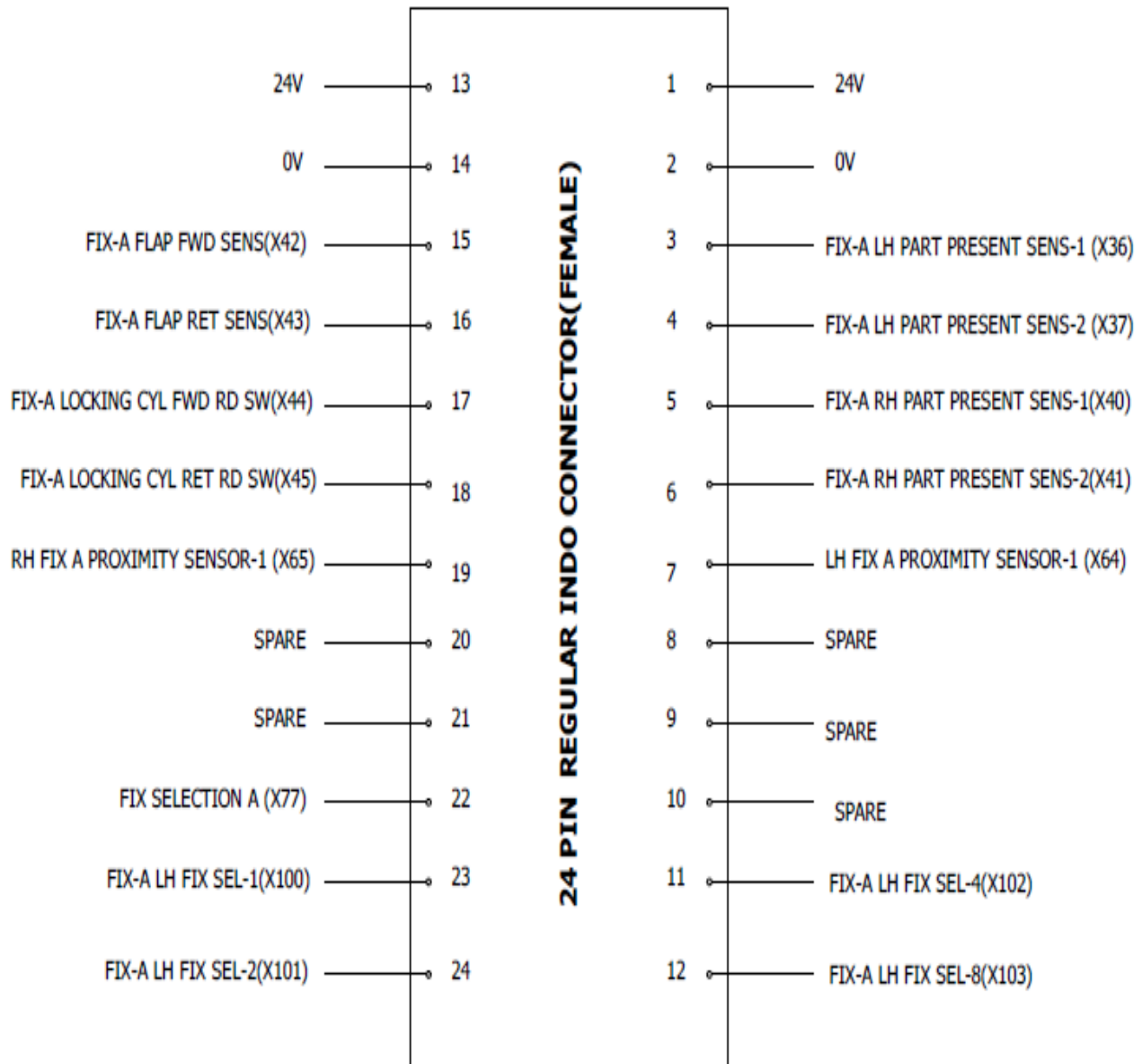
ENDPLATE

FROM SMPS
FROM SMPS
FIX-A LH PART PRESENT SENS-1
FIX-A LH PART PRESENT SENS-2
FIX-A RH PART PRESENT SENS-1
FIX-A RH PART PRESENT SENS-2
FIX-A FLAP FWD SENS
FIX-A FLAP RET SENS
FIX-A LOCKING CYL FWD RD SW
FIX-A LOCKING CYL RET RD SW
FIX-A LH FIX SEL-1
FIX-A LH FIX SEL-2
FIX-A LH FIX SEL-3
FIX-A LH FIX SEL-4
PALLATE B FIX FROM INDO
FROM SMPS
FROM SMPS
FIX-B LH PART PRESENT SENS-1
FIX-B LH PART PRESENT SENS-2
FIX-B RH PART PRESENT SENS-1
FIX-B RH PART PRESENT SENS-2
FIX-B RH FIX SEL-3
FIX-B RH FIX SEL-4
FIX-B FLAP FWD SENS
FIX-B FLAP RET SENS
FIX-B LOCKING CYL FWD RD SW
FIX-B LOCKING CYL RET RD SW
PALLATE A FIX FROM INDO
FIX-B RH FIX SEL-1
FIX-B RH FIX SEL-2

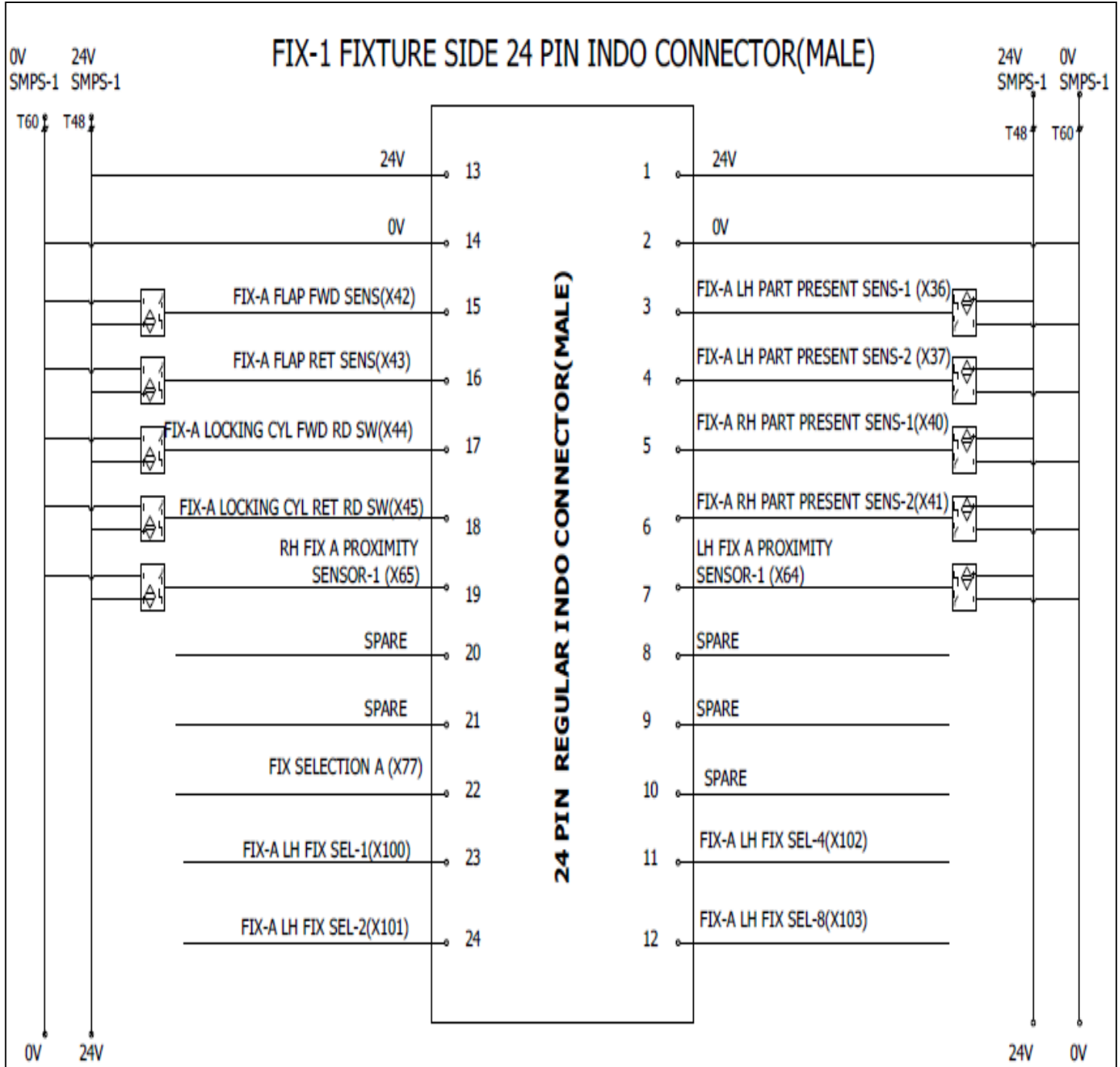
PANEL SIDE

FIX-1 MACHINE SIDE 24 PIN INDO CONNECTOR (FEMALE)

FIX-1 MACHINE SIDE 24 PIN INDO CONNECTOR(FEMALE)

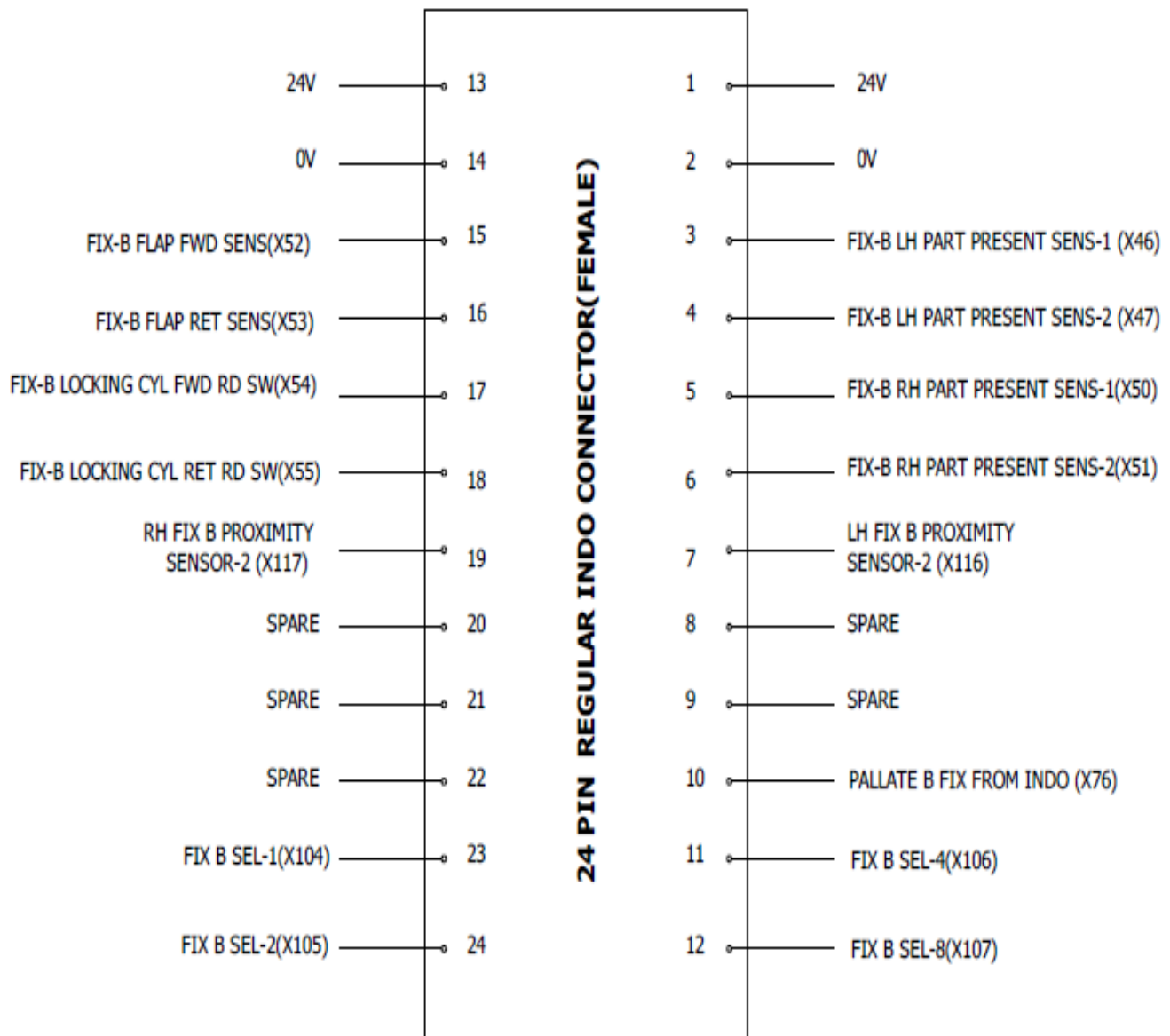


FIX-1 FIXTURE SIDE 24 PIN INDO CONNECTOR(FEMALE)

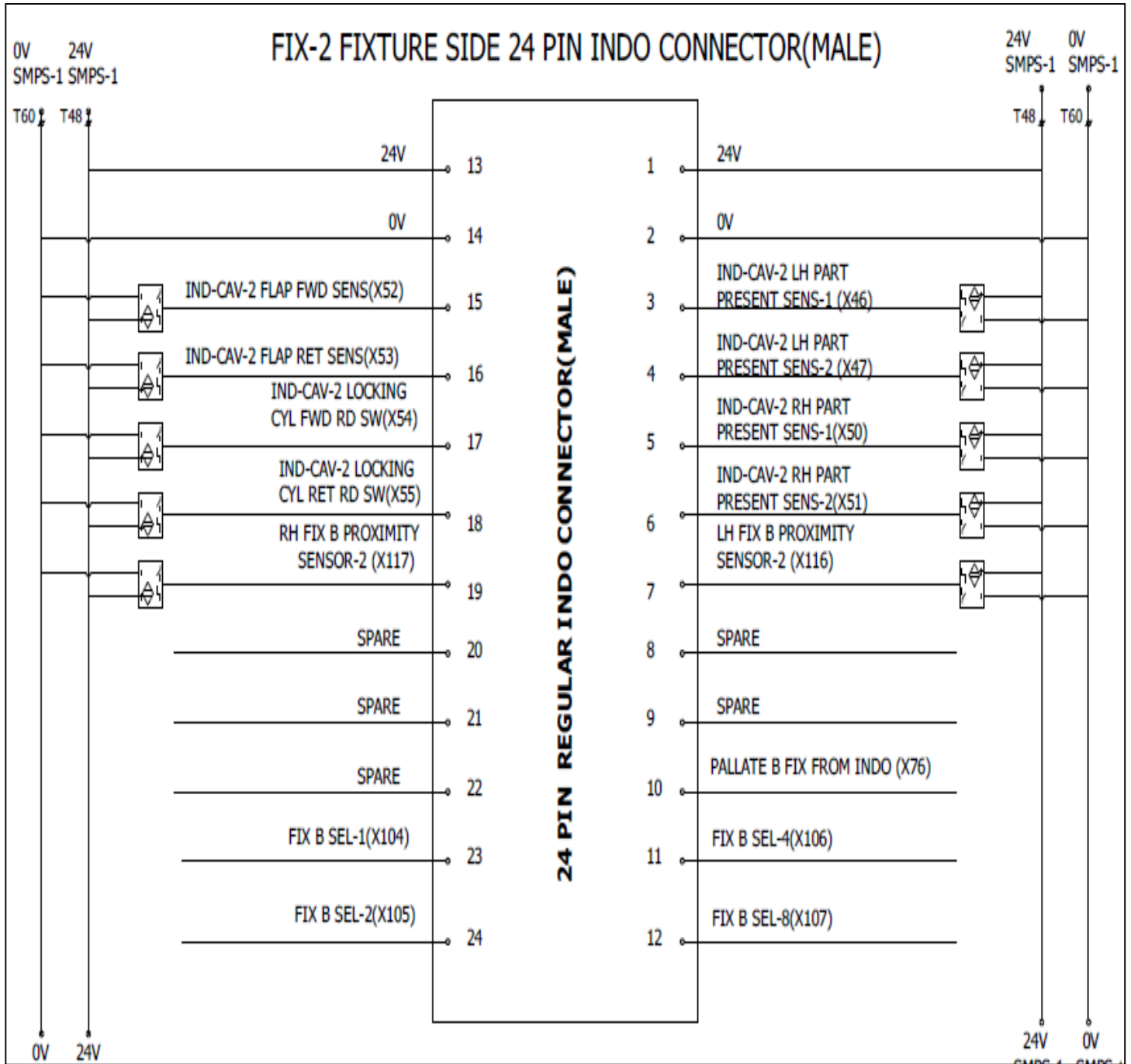


FIX-2 MACHINE SIDE 24 PIN INDO CONNECTOR (FEMALE)

FIX-2 MACHINE SIDE 24 PIN INDO CONNECTOR(FEMALE)

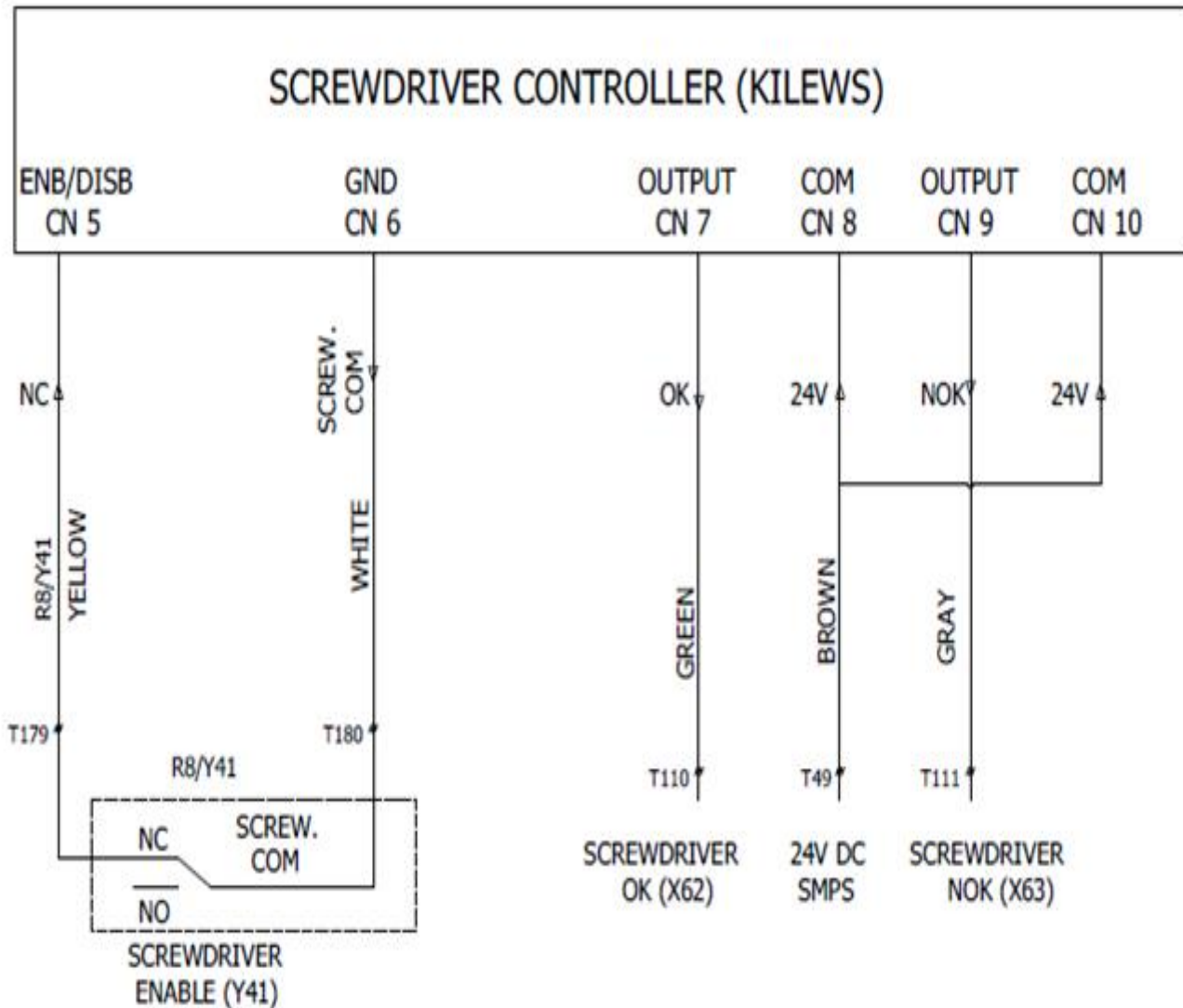


FIX-2 FIXTURE SIDE 24 PIN INDO CONNECTOR (FEMALE)



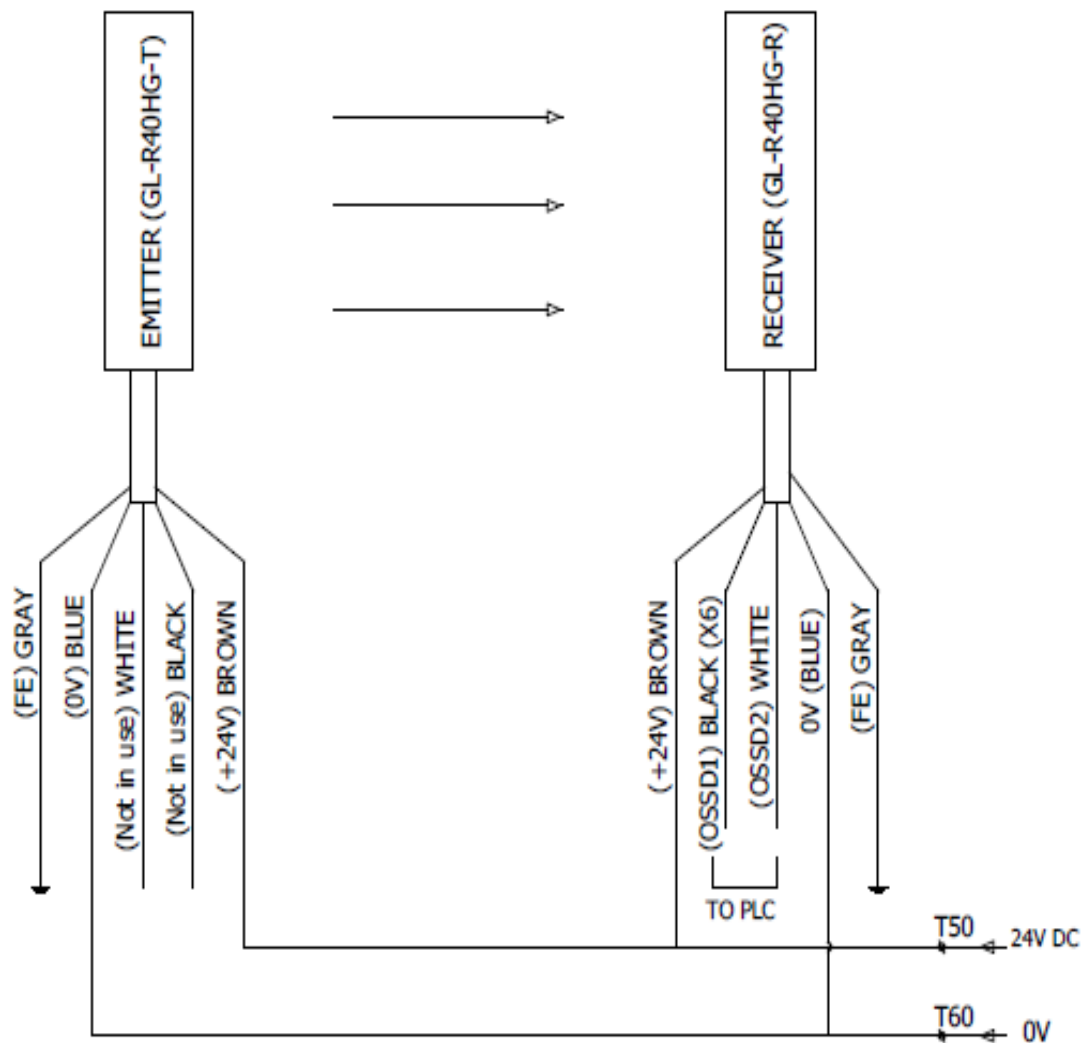
SCREWDRIVER CONNECTIONS (KELEIWS)

SCREWDRIVER CONNECTIONS (KILEWS)

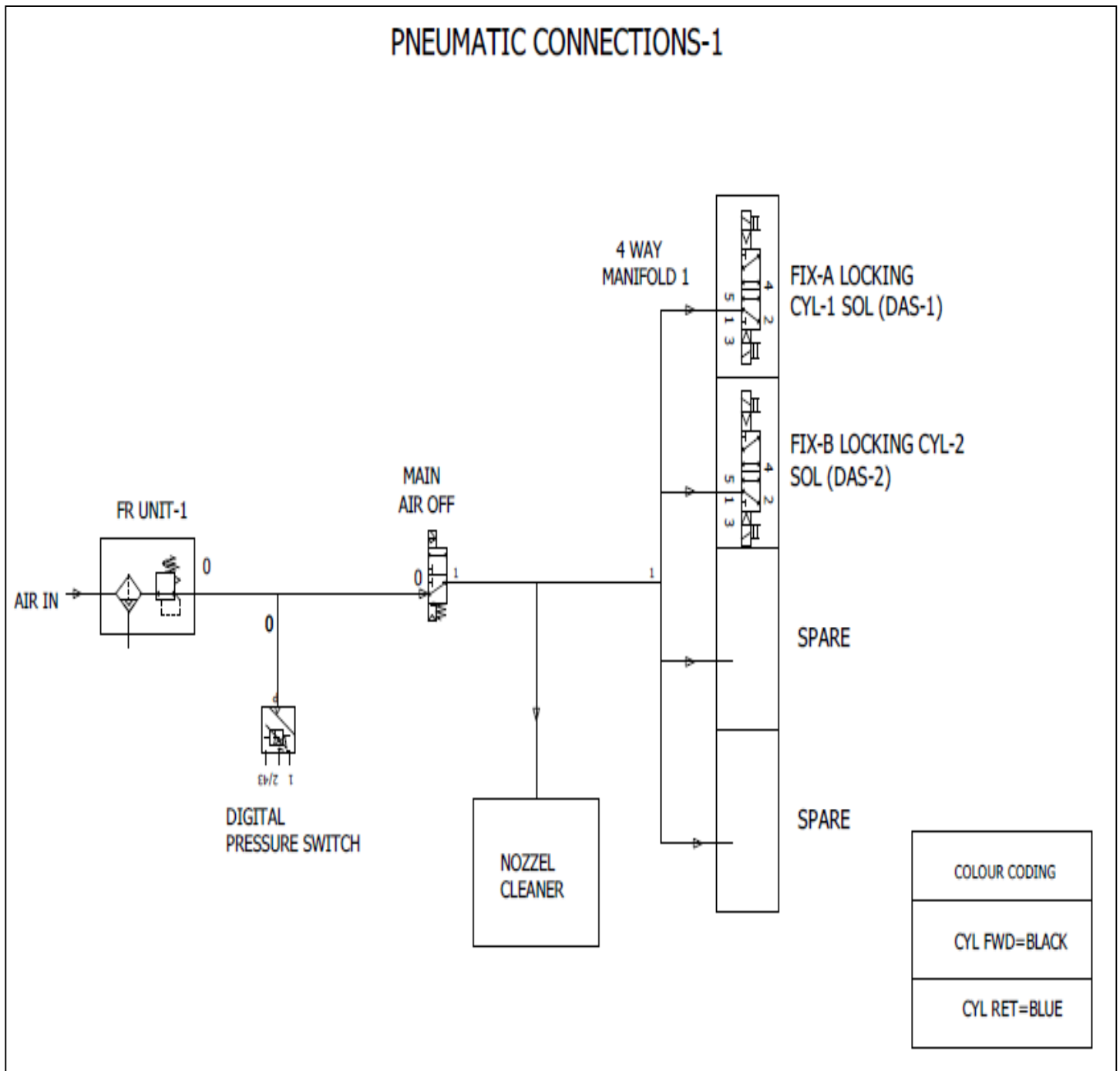


LIGHT CURTAIN CONNECTIONS

LIGHT CURTAIN CONNECTIONS (KEYENCE:-GL-R36HG)

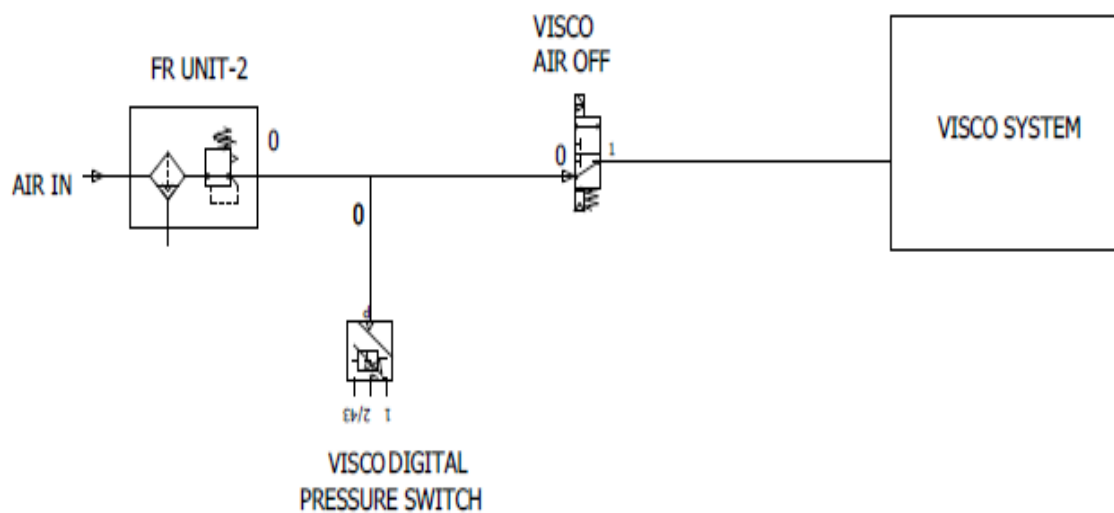


PNEUMATIC CONNECTIONS-1



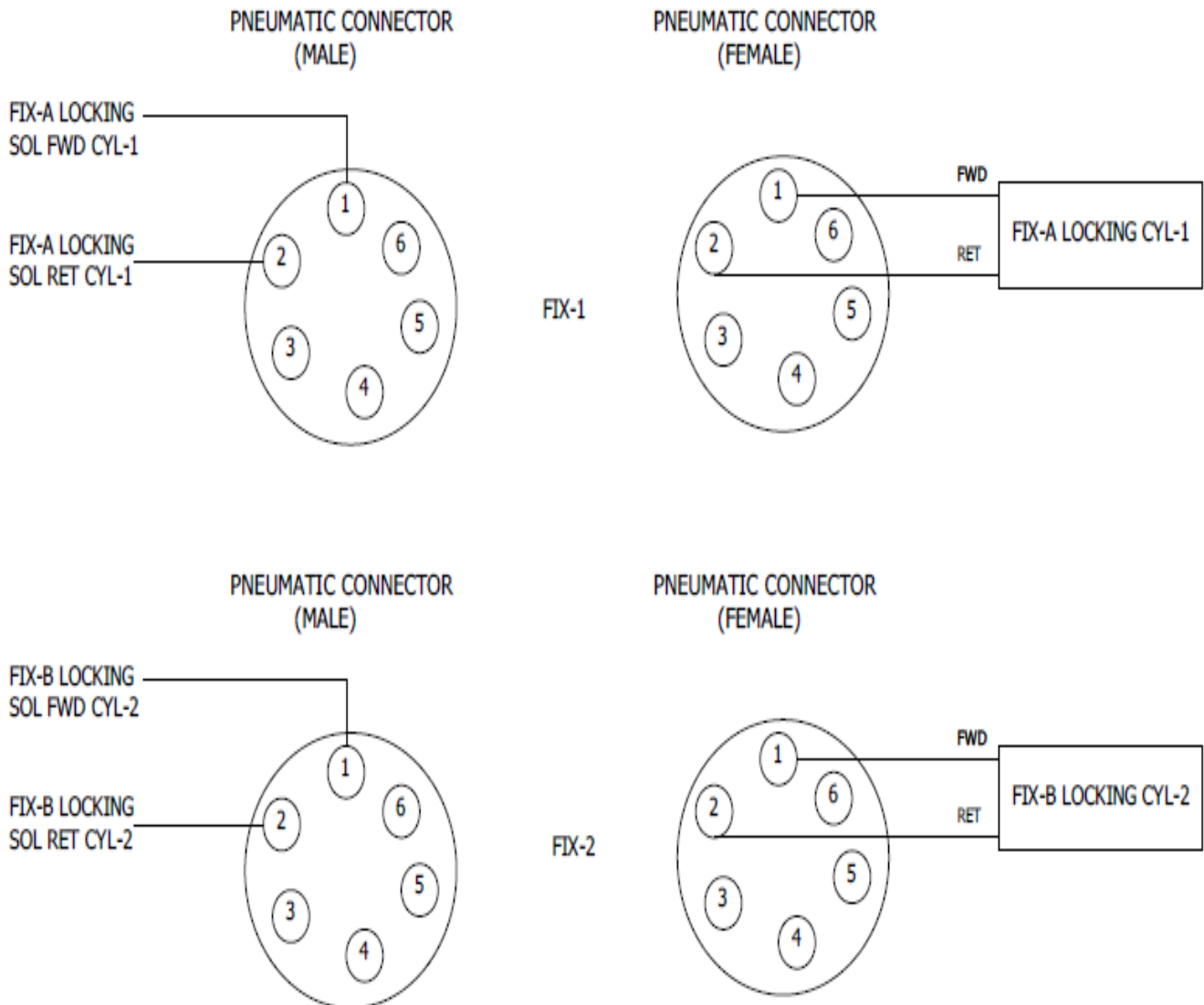
PNEUMATIC CONNECTIONS-2

PNEUMATIC CONNECTIONS-2



PNEUMATIC CONNECTOR (FIX-1 & 2)

PNEUMATIC CONNECTOR(FIX-1 AND 2)



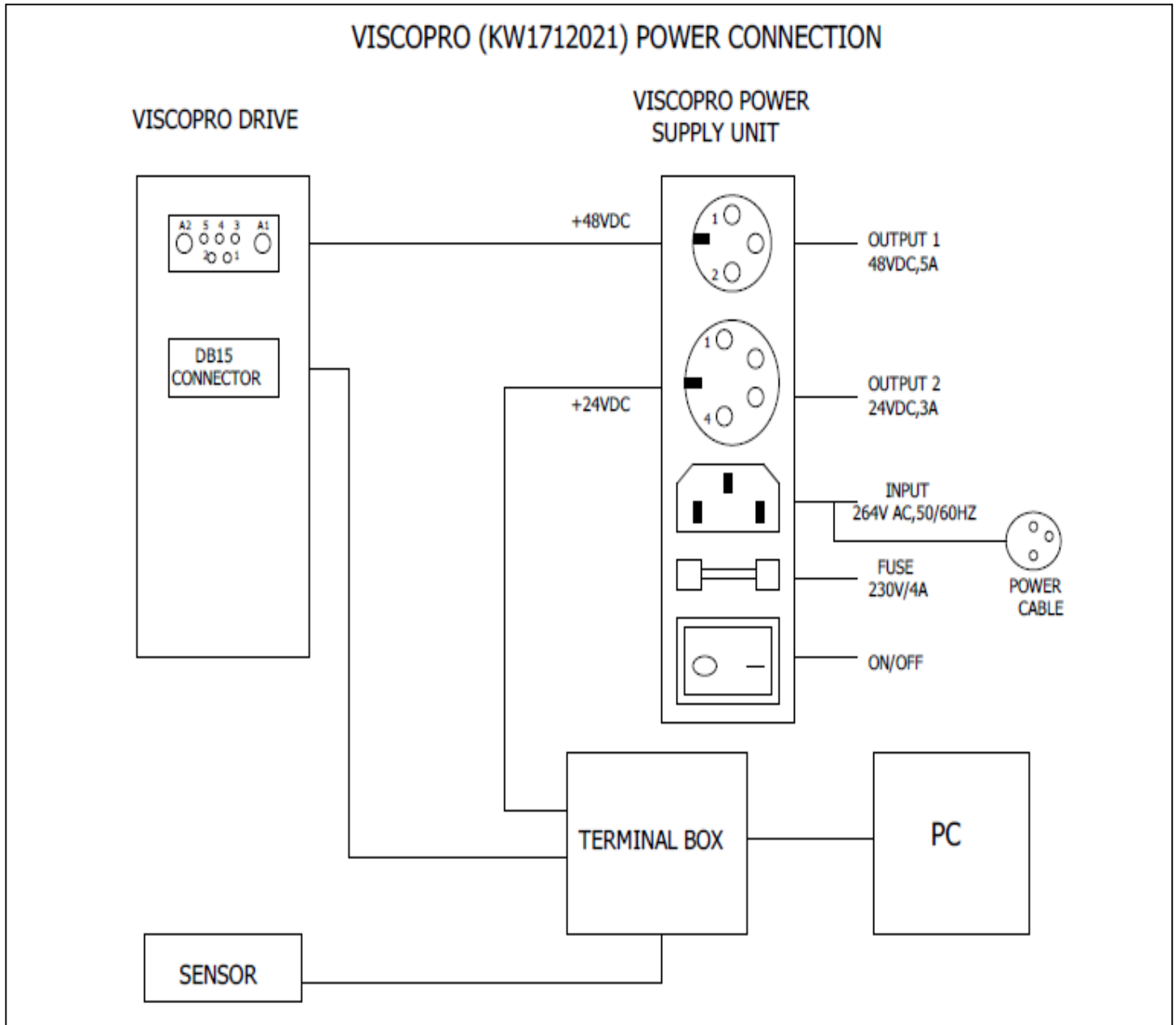
VISCO 10 CORE CABLE DETAILS

VISCO 10 CORE CABLE DETAILS

10 CORE CABLE NOS.	DESCRIPTION	VISCO CONTROLLER PIN NUMBERS
1	-	
2	VISCO START DISPENSING (Y32)	4
3	VISCO ERROR RESET(Y33)	6
4	VISCO(INPUT-3)	
5	SPARE	
6	VISCO INPUT PRESSURE OK(X30)	5
7	VISCO(CARTRIDGE NOT EMPTY)(X31)	7
8	VISCO READY(X32)	10
9	VISCO HEALTHY(X33)	12
10	VISCO SPARE(X34)	



VISCOPRO POWER CONNECTIONS



KEYENCE CAMERA CONNECTIONS DETAILS

KEYENCE CAMERA CONNECTIONS DETAILS

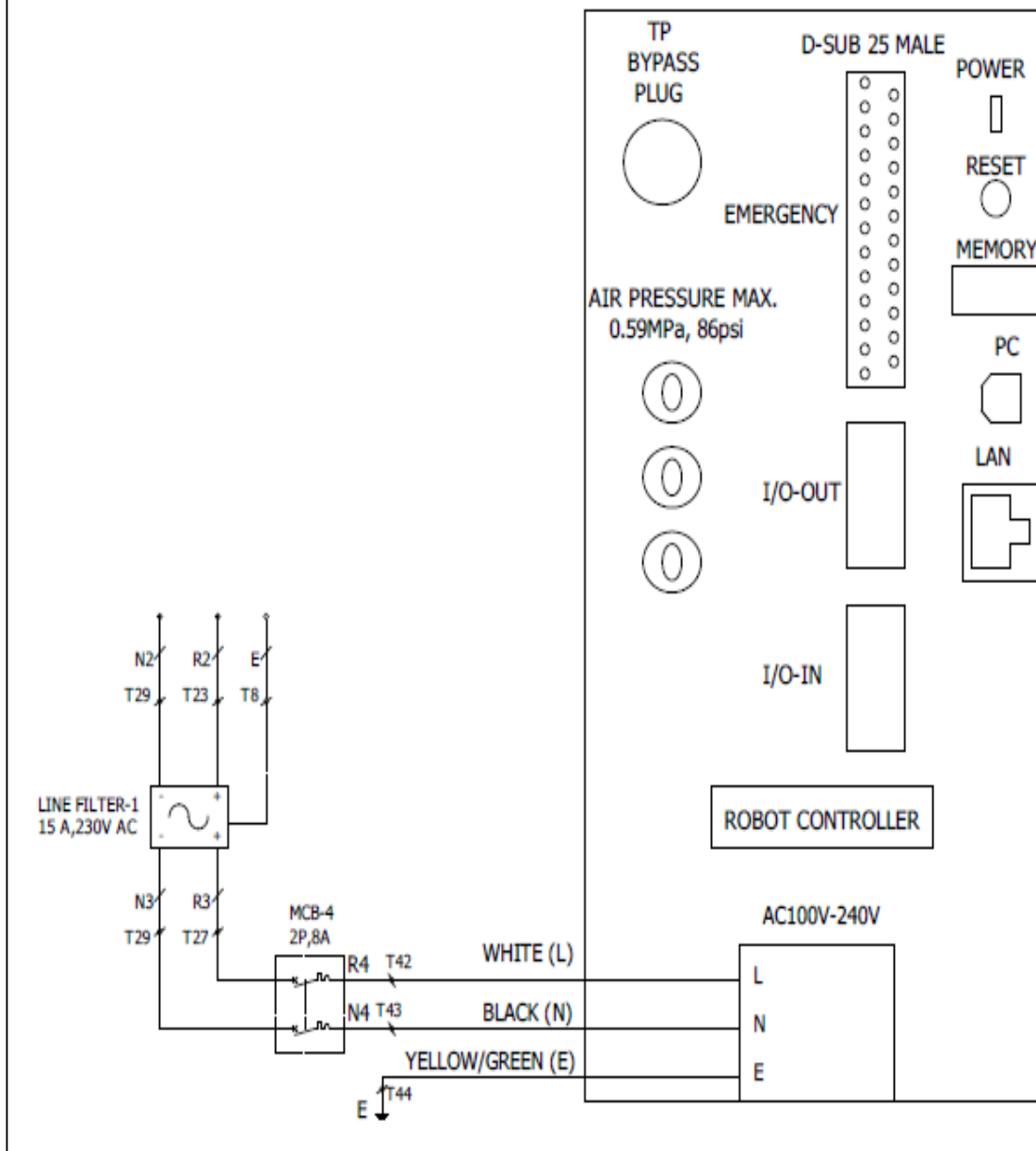
cables Details for IV2 Series (OP-87906)

PIN.NO.	WIRE COLOUR	NAME	PLC IO CAM-1	PLC IO CAM-2
A1	BROWN	IN1	Y60	Y67
A2	RED	IN2	Y61	Y70
A3	ORANGE	IN3	Y62	Y71
A4	YELLOW	IN4	Y63	Y72
A5	GREEN	IN5	Y64	Y73
A6	BLUE	IN6	Y65	Y74
A7	PURPLE	IN7	Y66	Y75
A8	GREY	IN8	-	-
A9	WHITE	UNUSED	-	-
A10	BLACK	UNUSED	-	-

PIN.NO.	WIRE COLOUR	NAME	PLC IO CAM-1	PLC IO CAM-2
B1	BROWN	OUT1	X70	X73
B2	RED	OUT2	X71	X74
B3	ORANGE	OUT3	X72	X75
B4	YELLOW	OUT4	X111	X113
B5	GREEN	OUT5	X112	X114
B6	BLUE	OUT6	-	-
B7	PURPLE	OUT7	-	-
B8	GREY	OUT8	-	-
B9	WHITE	UNUSED	-	-
B10	BLACK	UNUSED	-	-

SCARA ROBOT CONTROLLER DETAILS (T6-602S EPSON)

EPSON SCARA ROBOT CONTROLLER (T6-602S EPSON)



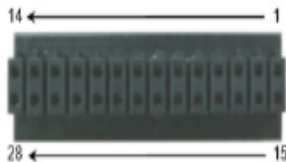
SCARA ROBOT I/O CABLE DETAILS (T6-602S EPSON)

SCARA ROBOT I/O CABLE DETAILS(T6-602S EPSON)

ROBOT INPUTS

PIN NO.	SIGNAL NAME	PLC ADDRESS
1	INPUT NO.-1(START)	Y20
15	INPUT NO.-2(PROG.SEL-1)	Y21
2	INPUT NO.-3(PROG.SEL-2)	Y22
16	INPUT NO.-4(PROG.SEL-4)	Y23
3	INPUT NO.-5(PROG.SEL-8)	Y24
17	INPUT NO.-6(STOP)	Y25
4	INPUT NO.-7(PAUSE)	Y26
18	INPUT NO.-8(CONTINUE)	Y27
5,19,14	INPUT COMMON NO. 0 TO 17	0V DC SMPS-1
6	INPUT NO.-9(RESET)	Y53
20	INPUT NO.-10(PROG.SELE-16)	Y54
7	INPUT NO.-11(FIX A INDICAT.)	Y55
21	INPUT NO.-12(MOVE NEXT POS)	Y56

I/O (Input) Connector pin assignment

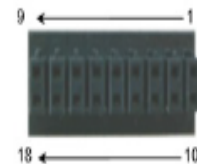


28 PIN INPUT CONNECTOR FOR ROBOT.

ROBOT OUTPUTS

PIN NO.	SIGNAL NAME	PLC ADDRESS
1	OUTPUT NO.-1(READY)	X20
10	OUTPUT NO.-2(RUNNING)	X21
2	OUTPUT NO.-3(PAUSE)	X22
11	OUTPUT NO.-4(ERROR)	X23
3	OUTPUT NO.-5(ESTOP ON)	X24
12	OUTPUT NO.-6(POS REACHED)	X25
4	OUTPUT NO.-7(DISPENSING ON)	X26
13	OUTPUT NO.-8(AT HOME POS)	X27
5	OUTPUT COMMON NO. 0 TO 7	24V DC SMPS-1

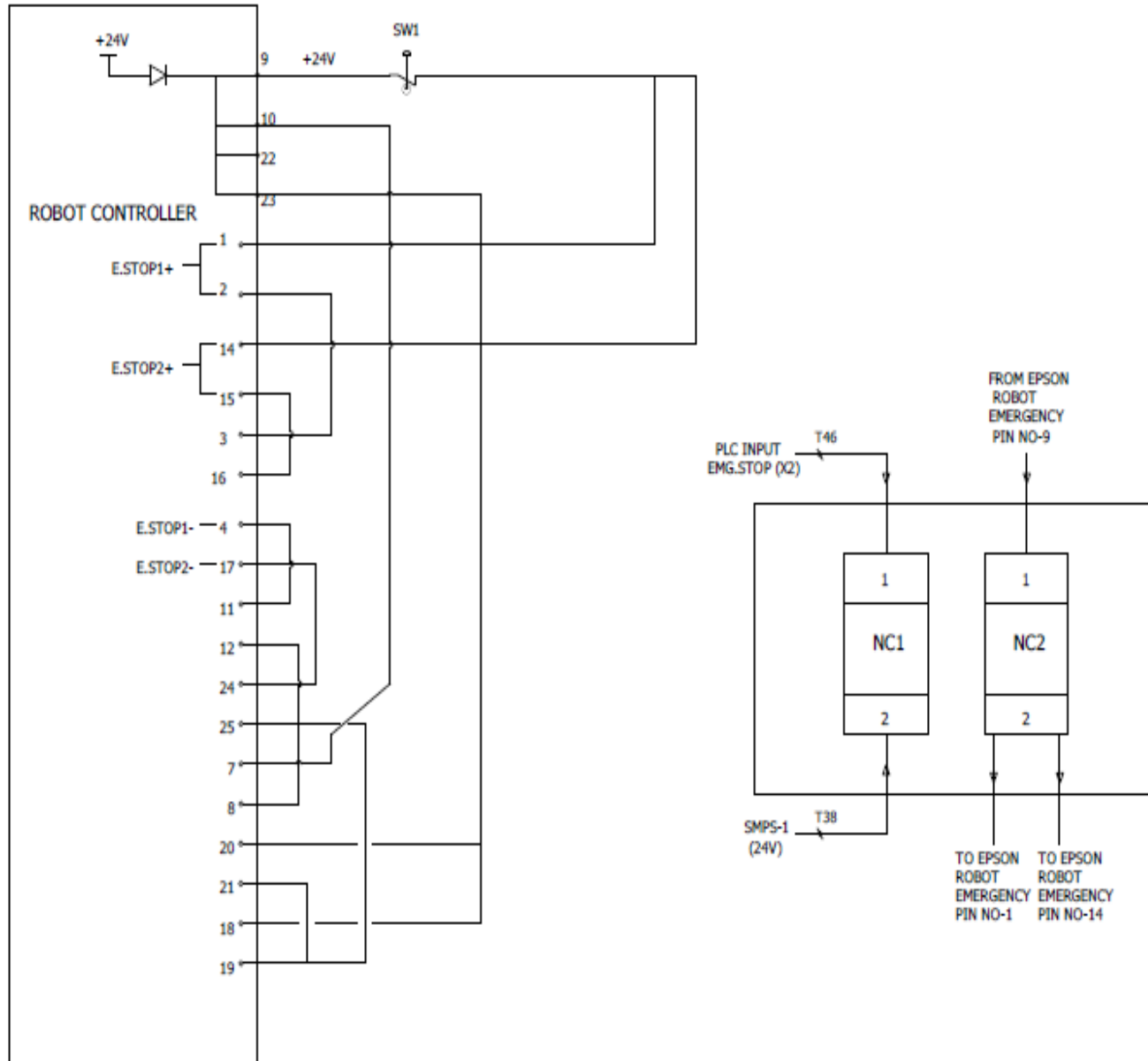
I/O (Output) Connector pin assignment



18 PIN OUTPUT CONNECTOR FOR ROBOT.

SCARA ROBOT EMERGENCY PB CONNECTIONS DETAILS(T6-602S EPSON)

EPSON SCARA ROBOT EMERGENCY PB CONNECTIONS DETAILS (T6-602S EPSON)



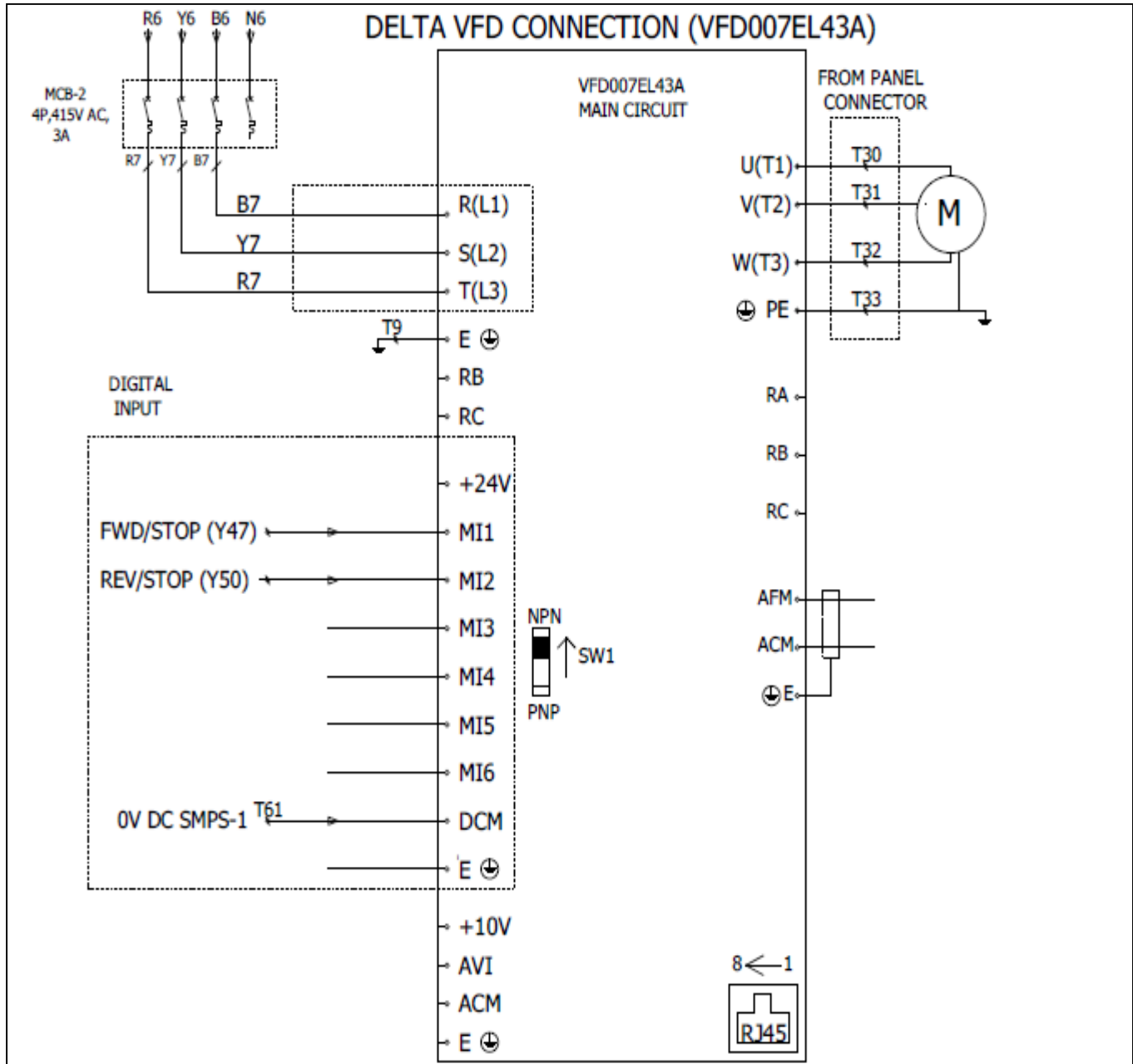
SCARA ROBOT EMERGENCY PIN CONNECTOR DETAILS(T6-602S EPSON)

SCARA ROBOT EMG STOP PIN CONNECTOR DETAILS (T6-602S EPSON)

ROBOT EMERGENCY PIN DESCRIPTION

PIN NO.	SIGNAL NAME	DESCRIPTION	PIN NO.	SIGNAL NAME	DESCRIPTION
1	ESW11	EMERGENCY STOP SWITCH CONTACT (1)	14	ESW21	EMERGENCY STOP SWITCH CONTACT (2)
2	ESW12	EMERGENCY STOP SWITCH CONTACT (1)	15	ESW22	EMERGENCY STOP SWITCH CONTACT (2)
3	ESTOP1+	EMERGENCY STOP CIRCUIT 1 (+)	16	ESTOP2+	EMERGENCY STOP CIRCUIT 2 (+)
4	ESTOP1-	EMERGENCY STOP CIRCUIT 1 (-)	17	ESTOP2-	EMERGENCY STOP CIRCUIT 2 (-)
5	NOT USED	-----	18	SDLATCH1	SAFEGUARD LATCH RELEASE
6	NOT USED	-----	19	SDLATCH2	SAFEGUARD LATCH RELEASE
7	SD11	SAFEGUARD INPUT (1)	20	SD21	SAFEGUARD INPUT (2)
8	SD12	SAFEGUARD INPUT (1)	21	SD22	SAFEGUARD INPUT (2)
9	24V	+24V OUTPUT	22	24V	+24V OUTPUT
10	24V	+24V OUTPUT	23	24V	+24V OUTPUT
11	24V GND	+24V GND OUTPUT	24	24V GND	+24V GND OUTPUT
12	24V GND	+24V GND OUTPUT	25	24V GND	+24V GND OUTPUT
13	NOT USED	-----			

VFD CONNECTIONS

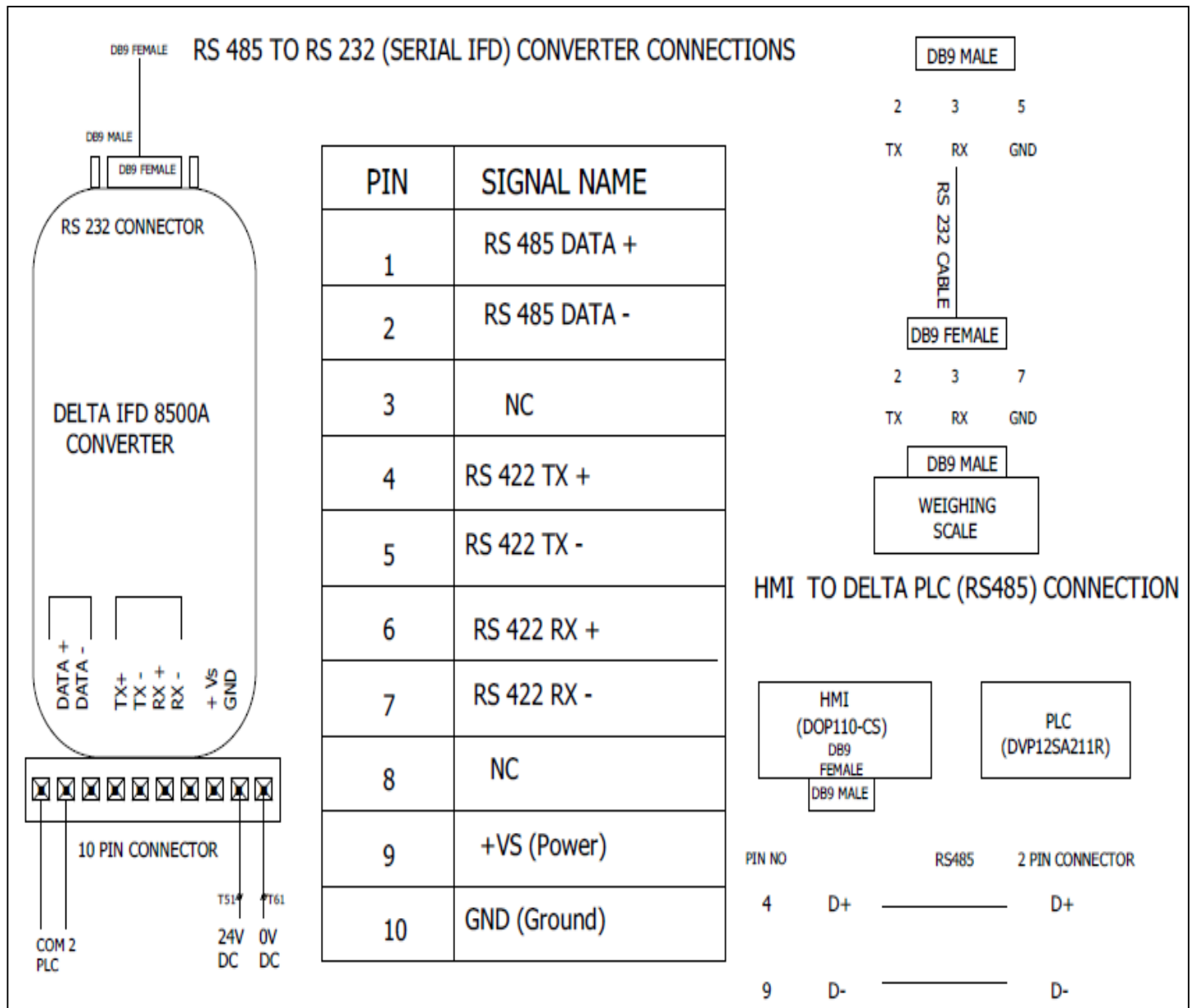


DELTA VFD PIN CONNECTOR DETAILS

DELTA VFD PIN CONNECTOR DETAILS (VFD007EL43A)

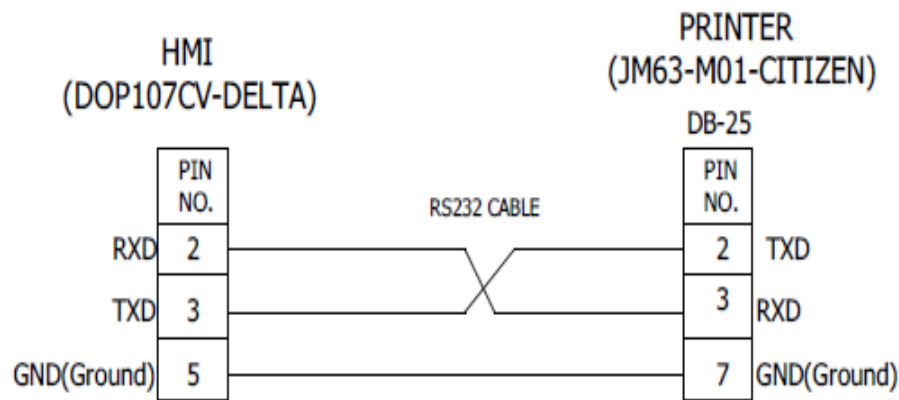
TERMINAL SYMBOL	TERMINAL FUNCTION	FACTORY SETTINGS (NPN MODE) ON:CONNECT TO DCM
MI1	FORWARD-STOP COMMAND	ON: RUN IN MI1 DIRECTION OFF: STOP ACC.TO STOP METNOD
MI2	REVERSE-STOP COMMAND	ON: RUN IN MI2 DIRECTION OFF: STOP ACC.TO STOP METNOD
MI3	MULTI-FUNCTION INPUT 3	REFER PROGRAMMING THE MULTI FUNCTION INPUTS
MI4	MULTI-FUNCTION INPUT 4	
MI5	MULTI-FUNCTION INPUT 5	ON: THE ACTIVATION CURRENT IS 5.5MA OFF: LEAKAGE CURRENT TOLERANCE IS 10 MicroA
MI6	MULTI-FUNCTION INPUT 6	
+24V	DC VOLTAGE SOURCE	+24VDC, 50MA USED FOR PNP MODE.
DCM	DIGITAL SIGNAL COMMON	COMMON FOR DIGITAL INPUTS AND USED FOR NPN MODE.
RA	MULTI-FUNCTION RELAY OUTPUT (N.O.) a	RESISTIVE LOAD
RB	MULTI-FUNCTION RELAY OUTPUT (N.C.) b	INDUCTIVE LOAD
Rc	MULTI-FUNCTION RELAY COMMON	
+10V	POTENTIOMETER POWER SUPPLY	+10VDC 3mA
AVI	ANALOG VOLTAGE INPUT	IMPEDANCE 47Kohm, RANGE: 0-10VDC,20MA
ACM	ANALOG CONTROL SIGNAL (COMMON)	COMMON FOR AVI= and AFM
AFM	ANALOG OUTPUT METER	0-10V, 2mA

RS 485 TO RS 232(SERIAL IFD) CONVERTER CONNECTIONS

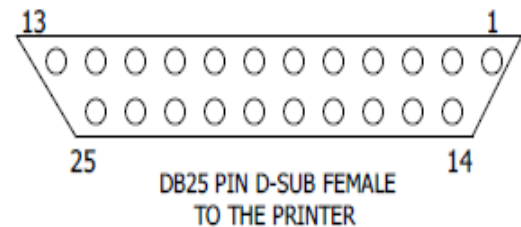


HMI TO RS232 PRINTER CONNECTIONS

HMI TO RS232 PRINTER(citizen JM-63-M01) CONNECTION



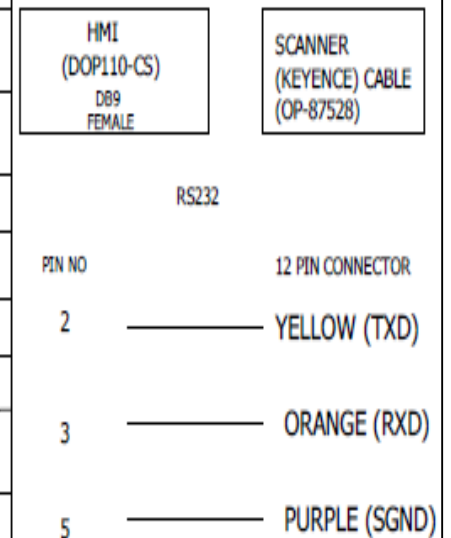
PIN	MODE-1		
	COM1	COM2	COM3
	RS-232	RS-232	RS-485
1	-	-	-
2	RXD	RXD	-
3	TXD	TXD	-
4	-	-	D+
5	GND		
6	-	-	-
7	RTS	RTS	-
8	CTS	CTS	-
9	-	-	D-



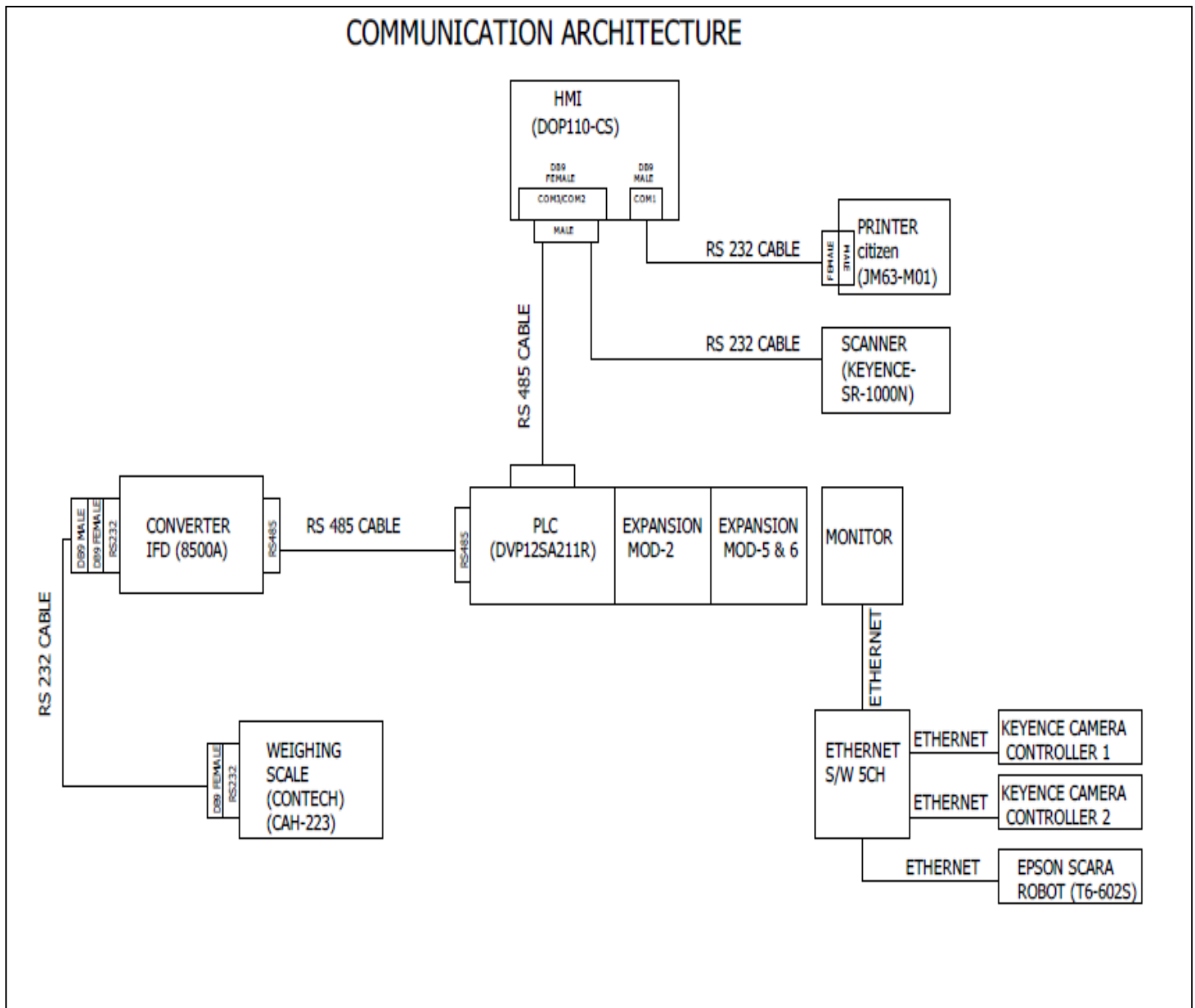
HMI TO SCANNER CONNECTIONS

DELTA HMI TO SCANNER KEYENCE CABLE (OP-87528) CONNECTION

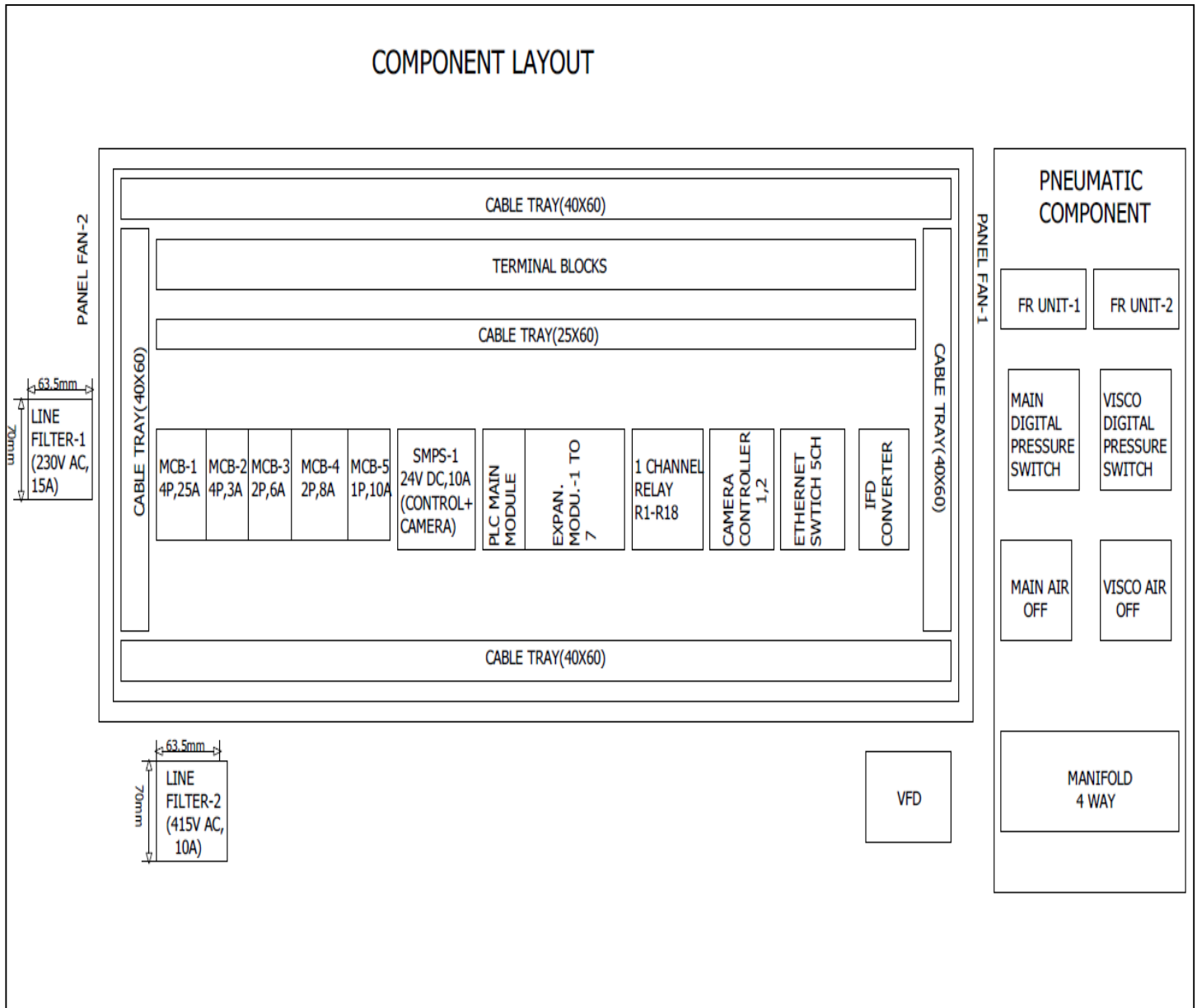
WIRE COLOR	NAME	DESCRIPTION	SIGNAL DIRECTION	INITIAL ASSIGNMENT
BROWN	24V	24VDC	INPUT	-----
BLUE	0V	POWER GND	-----	-----
ORANGE	RXD (RD)	RS-232C RECEIVE	INPUT	-----
YELLOW	TXD (SD)	RS-232C SEND	OUTPUT	-----
PURPLE	SGND	RS-232C GND	-----	-----
GREEN	IN1	INPUT SIGNAL 1 (Y57)	INPUT	READING START
GRAY	IN2	INPUT SIGNAL 2	INPUT	-----
WHITE AND BLUE	INCOM	INPUT COMMON (0V)	-----	-----
PINK	OUT1	OUTPUT SIGNAL 1 (X110)	OUTPUT	READING OK
LIGHT BLUE	OUT2	OUTPUT SIGNAL 2	OUTPUT	READING ERROR
WHITE	OUT3	OUTPUT SIGNAL 3	OUTPUT	BUSY
BLACK	OUT COM	OUTPUT COMMON (24V)	-----	-----
BLACK TUBE	FG	FRAME GROUND	-----	-----



COMMUNICATIONS ARCHITECTURE



COMPONENT LAYOUT



CABLE DETAILS

CABLE DETAILS				
CABLE	CORE DIAMETER	COLOUR CODE		MAKE
POWER CABLE	2.5 SQ MM 5 CORE FLEXIBLE	RED	PHASE-R	POLYCAB
		YELLOW	PHASE-Y	
		BLUE	PHASE-B	
		BLACK	NEUTRAL	
		YELLOW+GREEN	EARTH	
PANEL WIRING	CONTROL WIRING 0.5 SQ. MM,1 CORE	BLUE	+24V DC	RR CABLE
		WHITE	0V DC	
		GREY	I/P OF PLC	
		ORANEG	OUTPUT OF PLC	
	POWER WIRING 1.5 SQ. MM,3 CORE POWER WIRING 2 SQ. MM,3 CORE(FOR OUTPUT OF SMPS)	RED	LINE	
		BLACK	NEUTRAL	
		YELLOW+GREEN	EARTH	
TOWER LAMP	0.25 SQ.MM 6 CORE,SHIELDED	BLUE	0V DC	STANDARD
		GRAY/WHITE	SPARE	
		BROWN	BUZZER	
		GREEN	PASS LAMP	
		YELLOW	PROCESS LAMP	
		PINK	ERROR LAMP	
PLC TO HMI CABLE	RS 485 CABLE 2 CORE,0.25SQMM, SHIELDED	BROWN	D+	STANDARD
		WHITE	D-	

BILL OF MATERIAL:

YFG HL Thermal Glue Dispensing Machine					
Sr No	Component Name	Description	Part No.	Make	Qty
1	Control Panel	400 X 1000 X300	-	-	1
2	PLC	8DI/4DO	DVP12SA211R	Delta	1
3	Expansion Module	8DI/8DO	DVP16SP11R	Delta	6
4	Expansion Module	16DI	DVP16SM11N	Delta	1
5	HMI	10" Color	DOP110-CS	Delta	1
6	SMPS (control+robot+visco system)	24VDC,10A	S8VK-C24024	Omron	1
8	MCB (Mains)	4P,25A,415V AC	5SL64257RC	Siemens	1
9	MCB (SCARA ROBOT)	2P,8A ,230VAC	5SL62087RC	Siemens	1
10	MCB (CONTROL)	1P,10A ,24V DC	5SL61107RC	Siemens	1
11	MCB (CONTROL)	2P,6A ,230VAC	5SL62067RC	Siemens	1
13	MCB (VFD)	3P,3A ,415VAC	5SL4403-7RC	Siemens	1
14	Relay Board	1- Chanel	SOCY1-24D	connectwell	18
15	Line filter	230V AC,15A(control+robot)	EF-1B015B01E-C02	Elcom	1
		3 phase, 415V AC,10A(VFD)	EF-3P010D01E-C71	Elcom	1
16	ON/Off Switch	4P,25A	61194	LnT	1
17	PB Box	2Way	50mm	Teknic	2
18	Push Button (Green)	Mushroom Type	2AM3	Teknic	2
19	Reset Button(red)	Mushroom Type	2AM4	Teknic	1
20	Emergency Stop	Latched Type	2AML4	Teknic	1
21	NO Element	24 VDC	S1	Teknic	3
22	NC Element	24 VDC	S2	Teknic	1
23	Indication Lamp (Red)	230 VAC	3PLBR-4L	Teknic	1
24	Indication Lamp (Yellow)	230 VAC		Teknic	1
25	Indication Lamp (Blue)	230 VAC	3PLBR-6L	Teknic	1
26	Indication Lamp (Green)	230 VAC	3PLBR-3L	Teknic	1
27	Tower Lamp 3-Tier	Steady,24 VDC	MG-302B-RYG	Patellite	1
28	Filter	Ral7035	4"	Jainson	1
29	Fan (Control Panel)	230 VAC	REC 22038 A2T	Rexnord	2
30	Tube light (Work Station)	LED 230VAC,10W	LF101	SS Light	2
31	Tube light (Control Panel)	LED 230VAC,7W	MTL-2835-150-3W	SMC	1
32	Industrial socket	5 pin, 3 phase			1
33	Electrical quick connector	24 pin ,With bracket,(Male)	0110024M011	Indo Electric	2
		24 pin ,With bracket,(female)	0110024F011	Indo Electric	2
34	BARCODE SCANNER	Lector Professional 620(RS232	SR-1000	Keyence	1

		Comm.)			
35	VISION SYSTEM	24V DC- controller	IV2-G30F	Keyence	2
36	PC with monitor, mouse and Keyboard(2-LAN Card)			Dell	1
37	SCARA ROBOT	230V AC,5A,1.2KVA	T6-602S	EPSON	1
38	GLUING SYSTEM		VISCO PRO-C	VISCOTECH	1
39	RS485 to RS232 Converter	24V DC	IFD 8500A	DELTA	1
40	Printer	24 V DC	CL-S631II	CITIZEN	2
41	Sensors	24 V DC	Proxy		12
42	VFD	3 Phase,415V AC	VFD-EL	Delta	1
43	Light curtains	24V DC	GL-R36HG	Keyence	1
44	Screwdriver with controller	Input-115V/230V AC,50-60HZ,6.3A Output- DC 40V,9A,360W	KL-TCG	KILEWS	1
45	Limit switch	1NO/2NC contact	AZ16-12ZVRK-M16	Schmersal	2
46	Extension board	6A,230V AC	674 099	LEGREND	2
47	Weighing scale	220g,class-I,pan	size-90mm diameter	CONTECH	1
48	Ethernet switch	5 port Ethernet switch	SENB 5TX	Phoenix contact	1
49	Motor	415V AC 0.5HP,1.75A	1000BB2267	WHP	1

YFG HL Thermal Glue Dispensing Machine

Sr. No	Component Name	Description	Part No.	Make	Qty.
1	Solenoid Valve	Double Acting	SY7220-5LZ-02	SMC	2
2	Air Off Solenoid Valve	3-Port	VT307-5G1-02F	SMC	2
3	digital pressure switch	Pressure Display	ISE30AF-01-P-LA2	SMC	2
4	FRL unit	1/4"	AW20-02BG1	SMC	2
5	Pneumatic connector 6 pin	Male	DM6-06N-C2	SMC	2
		female	DM6-06N-C2	SMC	2
6	Reed switch	24v dc,3 wire	D-M9PL-LUG	SMC	4
7	Manifold	4-Way	SS5Y720-04	SMC	1
8	Manifold Plug	1/4"	PL-02	SMC	1
9	Straight Connector	1/4" X 6OD	KQ2H06-U02	SMC	1
10	Pneumatic Tube	Blue 6 OD	TU0604BU	SMC	5mtr.
		Black 6 OD	TU0604BK	SMC	5mtr.
11	Slip ring connector(electrical)	0~440VAC/VDC	MC5739-FL01	Moflon	1

I/O LIST:

YFG Thermal Glue Dispensing Machine					
	Inputs		Outputs		
PLC	ADDR	DESCRIPTION	ADDR	DESCRIPTION	RELAY
DVP12SA211R	X0	START PB1	Y0	PASS LAMP	
	X1	START PB 2	Y1	ERROR LAMP	
	X2	EMG STOP PB	Y2	PROCESS LAMP	
	X3	RESET PB	Y3	BUZZER	
	X4	MACHINE DIGITAL PSW	Y4	NA	
	X5	VISCO DIGITAL PSW	Y5	NA	
	X6	LIGHT CURTAIN FB	NA	NA	
	X7	REJECTION BIN SENSOR	NA	NA	
DVP-16SP-11R-1	X20	EPSON (READY)	Y20	EPSON(START)	
	X21	EPSON (RUNNING)	Y21	EPSON (PROG. SEL.-1)	
	X22	EPSON (PAUSE)	Y22	EPSON (PROG. SEL.-2)	
	X23	EPSON (ERROR)	Y23	EPSON (PROG. SEL.-4)	
	X24	EPSON (ESTOP ON)	Y24	EPSON (PROG. SEL.-8)	
	X25	EPSON (POS REACHED)	Y25	EPSON(STOP)	
	X26	EPSON (DISPENSING ON)	Y26	EPSON (PAUSE)	
	X27	EPSON (AT HOME POS)	Y27	EPSON (CONTINUE)	
DVP-16SP-11R-2	X30	VISCO(INPUT PRESSURE OK)	Y30	EPSON (SPARE)	
	X31	VISCO(CARTRIDGE NOT EMPTY)	Y31	EPSON (SPARE)	
	X32	VISCO(READY)	Y32	VISCO (START DISPENSING)	R1
	X33	VISCO(HEALTHY)	Y33	VISCO (ERROR RESET)	R2
	X34	VISCO(SPARE)	Y34	VISCO(INPUT-3)	R3
	X35	INDEXER HOME SENSOR	Y35	FIX-A LOCKING CYL FWD SOL	R4
	X36	FIX-A LH PART PRESENT SENS-1	Y36	FIX-A LOCKING CYL RET SOL	R5
	X37	FIX-A LH PART PRESENT SENS-2	Y37	FIX-B LOCKING CYL FWD SOL	R6
DVP-16SP-11R-3	X40	FIX-A RH PART PRESENT SENS-1	Y40	FIX-B LOCKING CYL RET SOL	R7
	X41	FIX-A RH PART PRESENT SENS-2	Y41	SCREWDRIVERR ENABLE	R8
	X42	FIX-A FLAP FWD SENS	Y42	SPARE	R9
	X43	FIX-A FLAP RET SENS	Y43	MAIN AIR OFF	R10
	X44	FIX-A LOCKING CYL FWD RD SW	Y44	VISCO AIR OFF	R11
	X45	FIX-A LOCKING CYL RET RD SW	Y45	PANEL FAN 1	R12
	X46	FIX-B LH PART PRESENT SENS-1	Y46	PANEL FAN 2	R13

	X47	FIX-B LH PART PRESENT SENS-2	Y47	VFD STATR CW	R14
DVP-16SP-11R-4	X50	FIX-B RH PART PRESENT SENS-1	Y50	VFD START CCW	R15
	X51	FIX-B RH PART PRESENT SENS-2	Y51	VFD SPEED_SELECT	R16
	X52	FIX-B FLAP FWD SENS	Y52	VFD BREAK	R17
	X53	FIX-B FLAP RET SENS	Y53	EPSON RESET	R18
	X54	FIX-B LOCKING CYL FWD RD SW	Y54	EPSON PROGM SEL.16	
	X55	FIX-B LOCKING CYL RET RD SW	Y55	EPSON FIX A INDICATION	
	X56	HOMING SENSOR-1	Y56	EPSON MOVE TO NEXT POS	
	X57	HOMING SENSOR-2	Y57	SICK SCANNER TRIGGER	
DVP-16SP-11R-5	X60	DOOR LOCK SWITCH	Y60	RH TRIGGER	
	X61	DOOR LOCK SWITCH	Y61	RH CAM SCENE SEL-1	
	X62	SCREWDRIVER OK	Y62	RH CAM SCENE SEL-2	
	X63	SCREWDRIVER NOK	Y63	RH CAM SCENE SEL-4	
	X64	VFD ERROR	Y64	RH CAM CMD LOW	
	X65	VFD RUNNING	Y65	RH CAM CMD HIGH	
	X66	FIX-B AT OPERATOR POS	Y66	CAM-1 EXCECUTE	
	X67	FIX-A AT OPERATOR POS	Y67	LH CAM TRIGGER	
DVP-16SP-11R-6	X70	CAM-1 OK(OUT-1)	Y70	LH CAM SCENE SEL-1	
	X71	CAM-1 RUN/BUSY(OUT-2)	Y71	LH CAM SCENE SEL-2	
	X72	CAM-1 NOK/Error(OUT-3)	Y72	LH CAM SCENE SEL-4	
	X73	CAM-2 OK(OUT-1)	Y73	LH CAM CMD LOW	
	X74	CAM-2 RUN/BUSY(OUT-2)	Y74	LH CAM CMD HIGH	
	X75	CAM-2 NOK/Error(OUT-3)	Y75	LH CAM EXCECUTE	
	X76	PALLATE A FIX FROM INDO	Y76	SPARE	
	X77	PALLATE B FIX FROM INDO	Y77	SPARE	
DVP-16SM-11N-7	X100	FIX-A FIX SEL-1			
	X101	FIX-A FIX SEL-2			
	X102	FIX-A FIX SEL-4			
	X103	FIX-A FIX SEL-8			
	X104	FIX-B FIX SEL-1			
	X105	FIX-B FIX SEL-2			
	X106	FIX-B FIX SEL-4			
	X107	FIX-B FIX SEL-8			
	X110	SCANNER (OUT 1) OK			
	X111	CAM_1 (OUT 4)			
	X112	CAM_1 (OUT 5)			
	X113	CAM_2 (OUT 4)			
	X114	CAM_2 (OUT 5)			

**YFG HL THERMAL GLUE DISPENSING MACHINE**

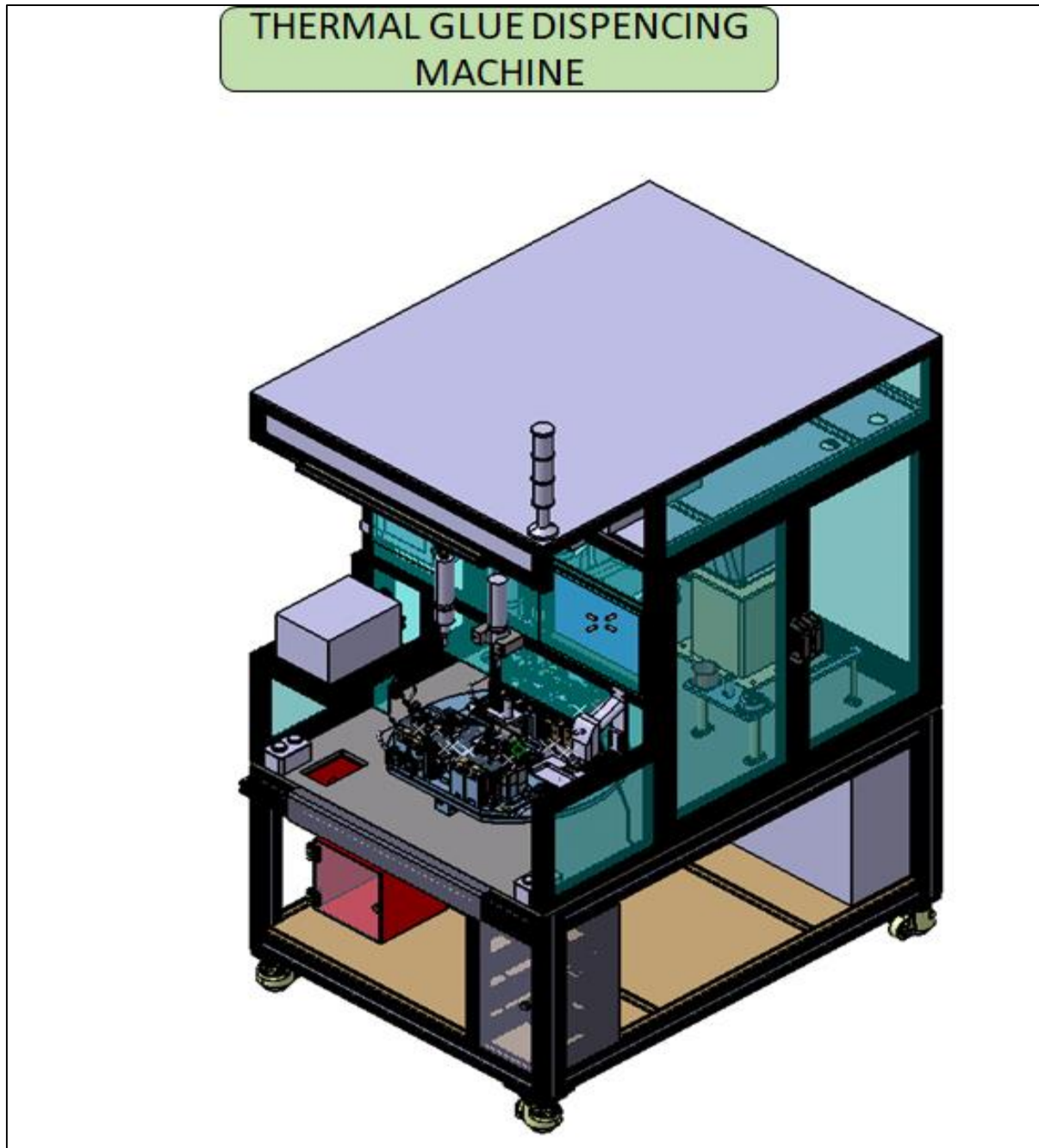
MARELLI MOTHERSON AUTO SYSTEM LTD.
GAT NR. 148-150, VILLAGE AMBETHAN TALUKA KHED,
NEAR DWARKA SCHOOL, PUNE, MAHARASHTRA- 410501



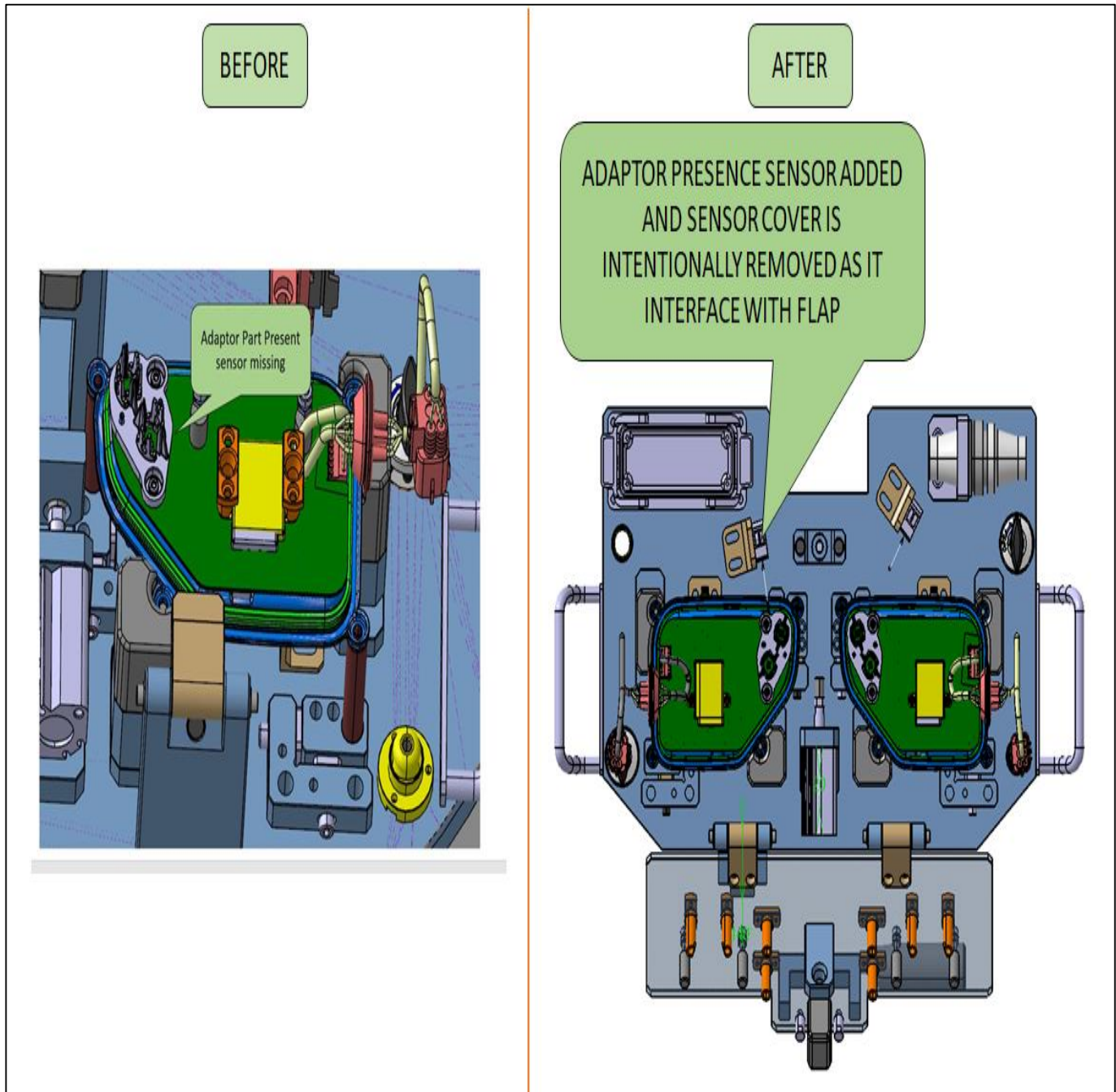
X115	SPARE			
X116	LH FIX B PROXIMITY SENSOR-2			
X117	RH FIX B PROXIMITY SENSOR-2			

WORKSTATION DETAILS:

THERMAL GLUE DISPENSING MACHINE

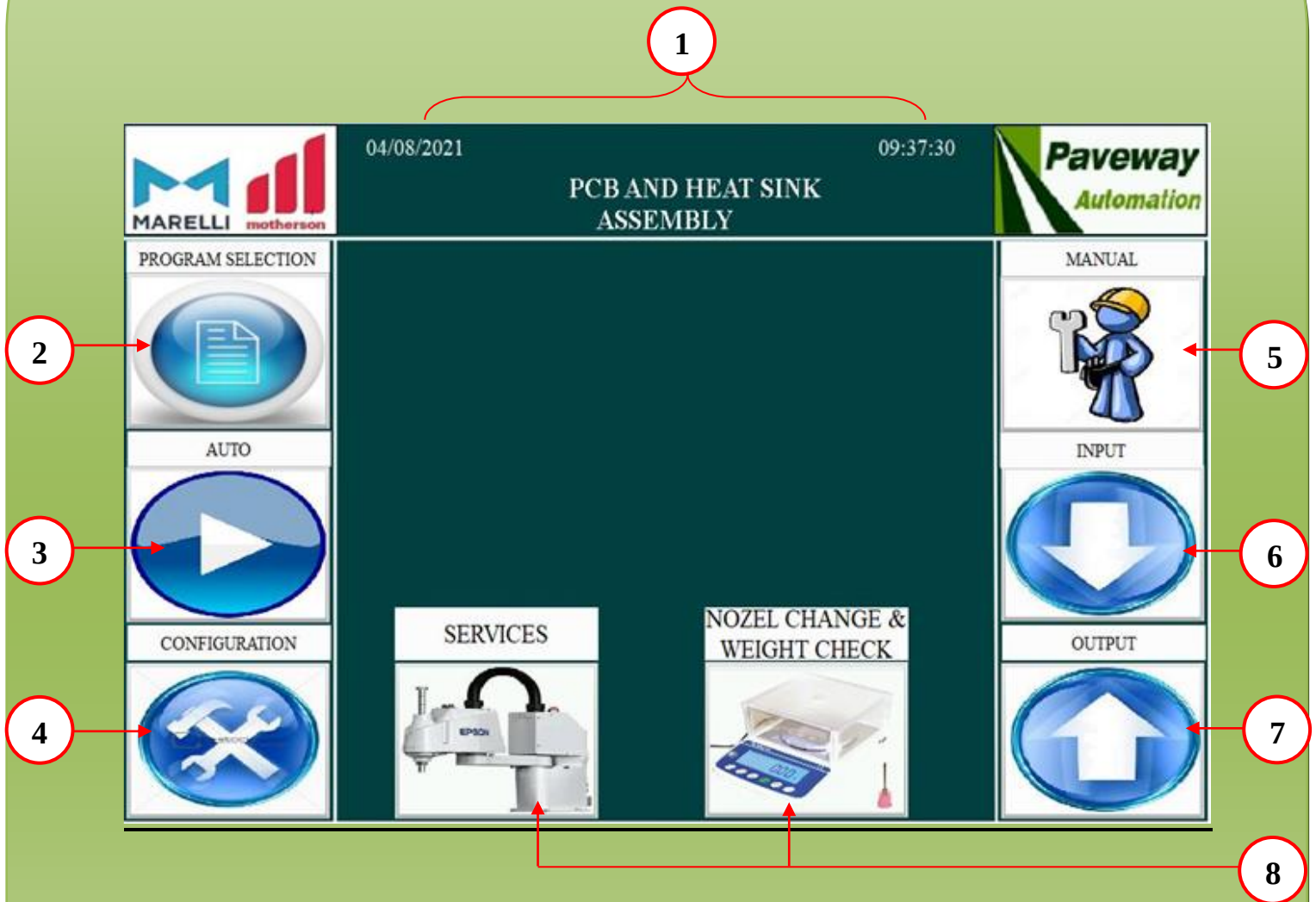


YFG HL FIXTURE DETAILS:



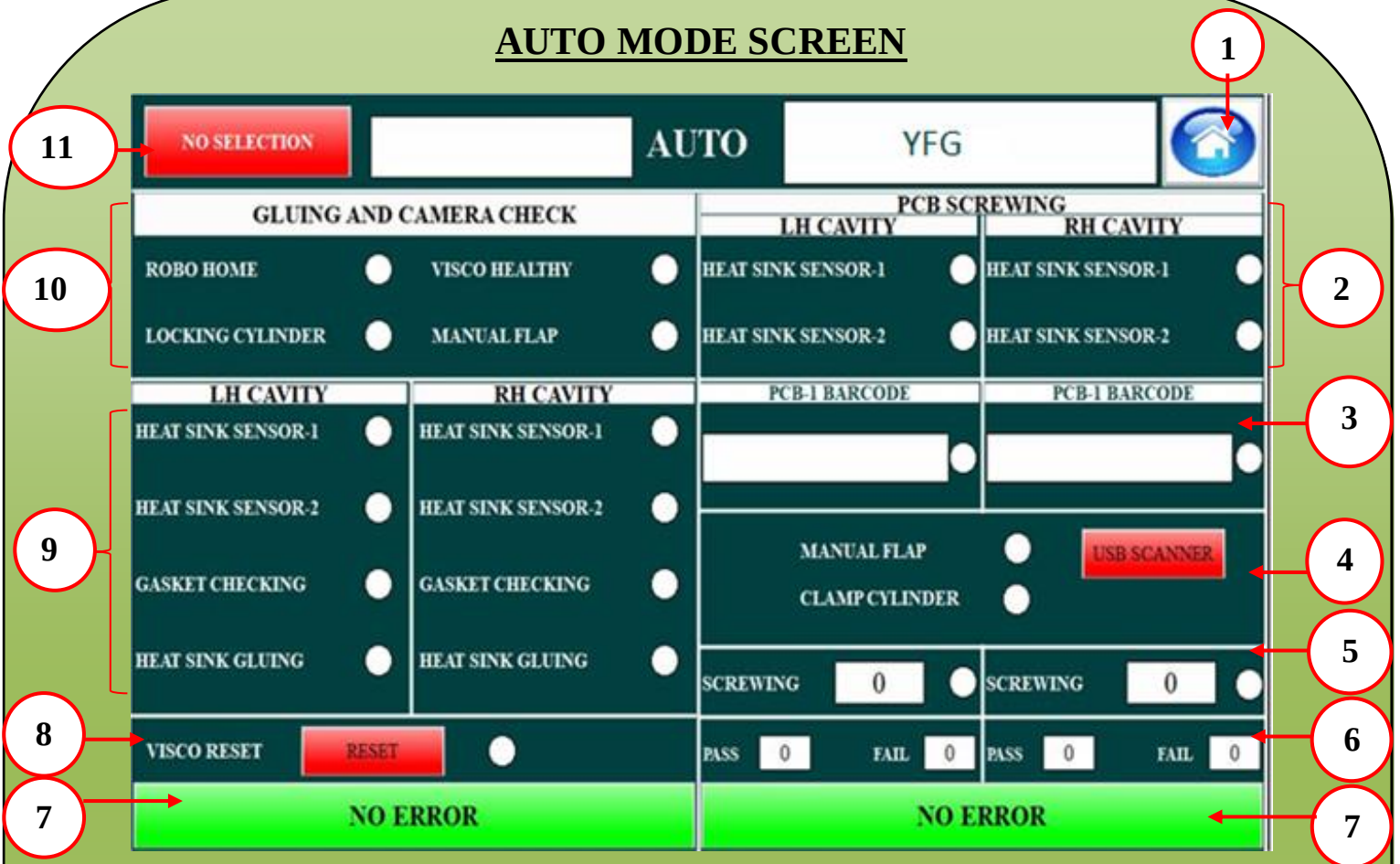
HMI SCREEN DETAILS:-

HOME SCREEN



- 1.Date & time display.
- 2.Programs screen navigation button
- 3.Auto screen navigation button
- 4.configuration screen navigation button.
- 5.Manual screen navigation button.
- 6.Input screen navigation button.
- 7.Output screen navigation button.
- 8.Services,nozel change & weight check

AUTO MODE SCREEN



1. Home screen navigation option.
2. PCB screwing LH & RH cavity sensor status.
3. LH & RH cavity PCB-1 barcode string display & status.
4. Manual flap, clamp cylinder status & USB scanner button.
5. PCB screwing LH & RH cavity screwing count display and status.
6. PCB screwing LH & RH cavity pass, fail count display.
7. Status message display.
8. Visco system reset button & status.

9.LH & RH cavity sensor status.

10.Robot home position,visco healthy,locking cylinder & manual flap status.

11.LH & RH Navigation screen button, display which one one is selected.

PURGE AND NOZEL CHANGE SETTING SCREEN

1

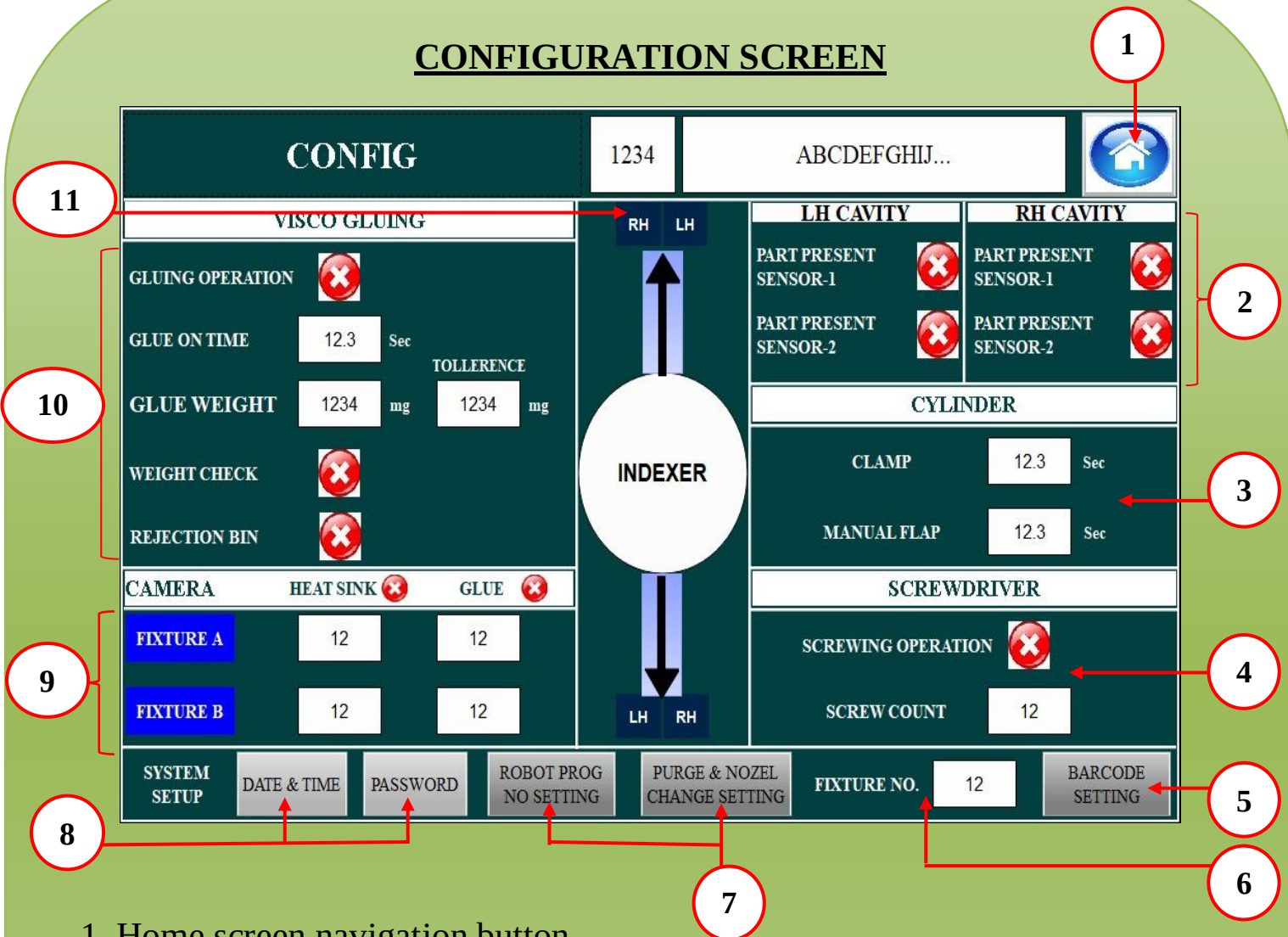
PURGE AND NOZEL CHANGE SETTINGS



PURGE CYCLE START TIMER	0.0 Sec	2
AUTO OIL DIP TIMER	0.0 Sec	3
NO. OF CYCLE FOR NOZEL CENTER CHECK	0 No.	4
NO. OF CYCLE FOR NOZEL CHANGE	0 No.	5
PURGE ON TIME	0.0 Sec	6
NOZEL CLEAN ON TIME	0.0 Sec	7

1. Home screen navigation button.
2. Purge cycle start timer time display.
3. Auto oil dip timer time display.
4. Number of cycle for nozzle center check count display.
5. Number of cycle for nozzle change count display.
6. Purge on time display.
7. Nozzel clean on time display.

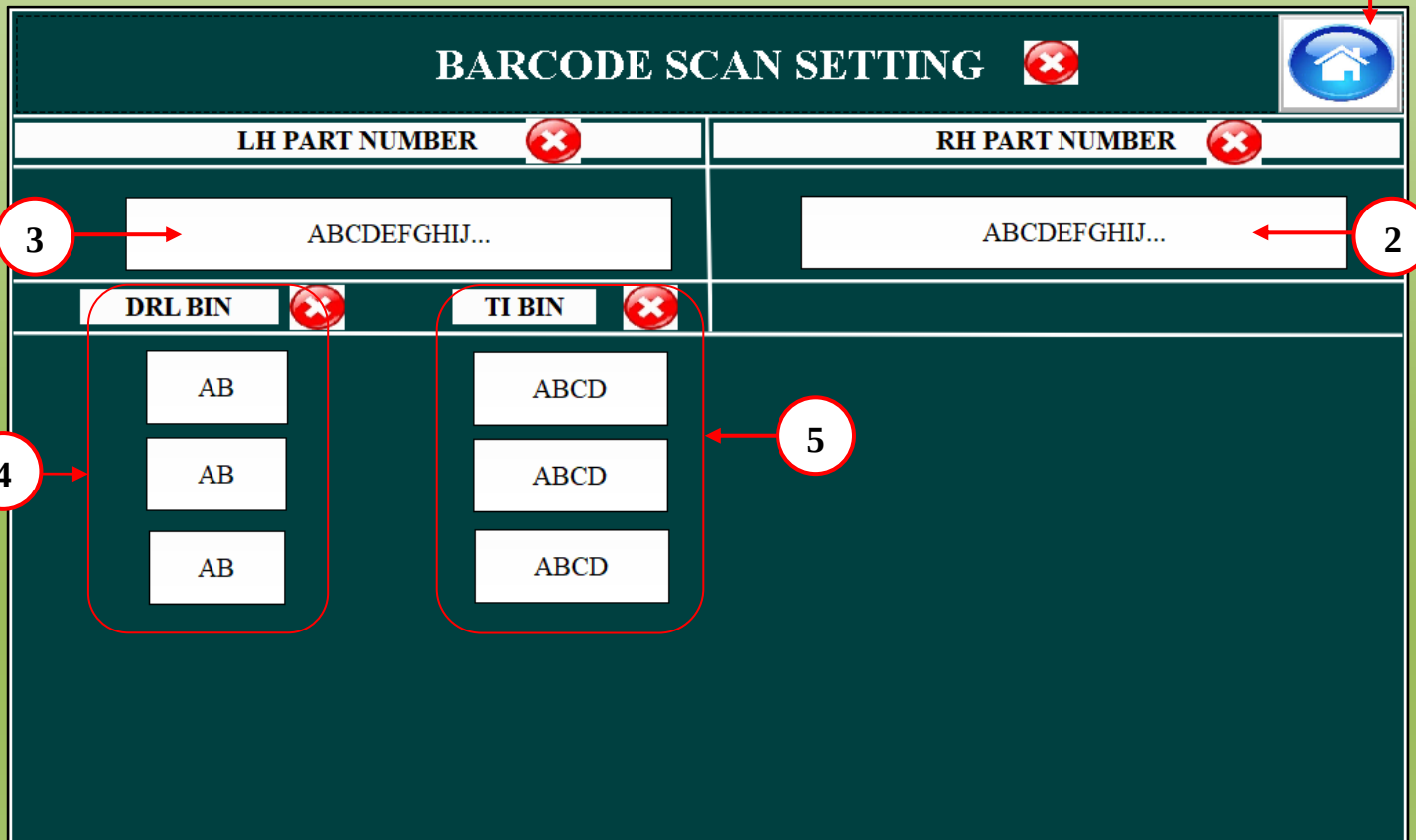
CONFIGURATION SCREEN



1. Home screen navigation button.
2. LH & RH cavity sensors select/unselect option.
3. Clamp cylinder & manual flap operating time.
4. Screwing operation select/unselect & count setting.
5. Barcode setting navigation button.
6. Fixture number setting.
7. Robot program number setting & purge, nozzle change setting buttons.
8. Date, time & password setting buttons.

9. Parameter setting for fixture A and B.
10. Visco gluing parameter setting.
11. Indexer parameter setting.

BARCODE SCAN SETTING



The screenshot shows the 'BARCODE SCAN SETTING' screen. At the top right is a home icon (1). Below it are two input fields: 'LH PART NUMBER' (3) and 'RH PART NUMBER' (2), both with red 'X' icons. Below these are two sections: 'DRL BIN' (4) and 'TI BIN' (5), each with a red 'X' icon. The 'DRL BIN' section contains three 'AB' labels. The 'TI BIN' section contains three 'ABCD' labels. Red arrows and circles with numbers 1 through 5 point to these specific elements.

1. Home screen navigation button.
2. RH part number setting.
3. LH part number setting.
4. DRL bin Color setting.
5. TI bin Color setting.

MANUAL SCREEN

PALLET-A				PALLET-B			
LH CAVITY		RH CAVITY		LH CAVITY		RH CAVITY	
PART PRESENT SENSOR	☐	PART PRESENT SENSOR	☐	PART PRESENT SENSOR	☐	PART PRESENT SENSOR	☐
CLAMP CYLINDER	RET	FWD	RET	CLAMP CYLINDER	RET	FWD	RET
MANUAL FLAP				MANUAL FLAP			
VISCO GLUING				SCARA ROBOT			
VISCO READY	☐	VISCO HEALTHY	☐	LH GLUE POS-1	GO	RH GLUE POS-1	GO
VISCO CARTRIDGE STATUS	☐	VISCO INLET PRESS	☐	LH GLUE POS-2	GO	RH GLUE POS-2	GO
GLUE OPERATION	START	0.0	Sec	HOMING	GO	NOZEL CHANGE POS	GO
VISCO RESET	RESET			PURGE POS	GO	CENTER CHECK POS	GO
RH CAMERA		LH CAMERA		NOZEL CLEAN POS	GO	OIL DIP POS	GO
SCENE NO.	0	SCENE NO.	0				
TRIGGER	OK	OK	NOK				
	☐		☐				
SCREWDRIVER	ENABLE	OK	NOK	BUZZER OFF		INDEX	START
	☐		☐				HOME
							☐

1. Home screen navigation button.

2. Pallate-B LH and RH cavity sensors status.

3. Pallate-B Clamp cylinder and manual flap reed switch status, clamp cylinder fwd/ret manual operation button.

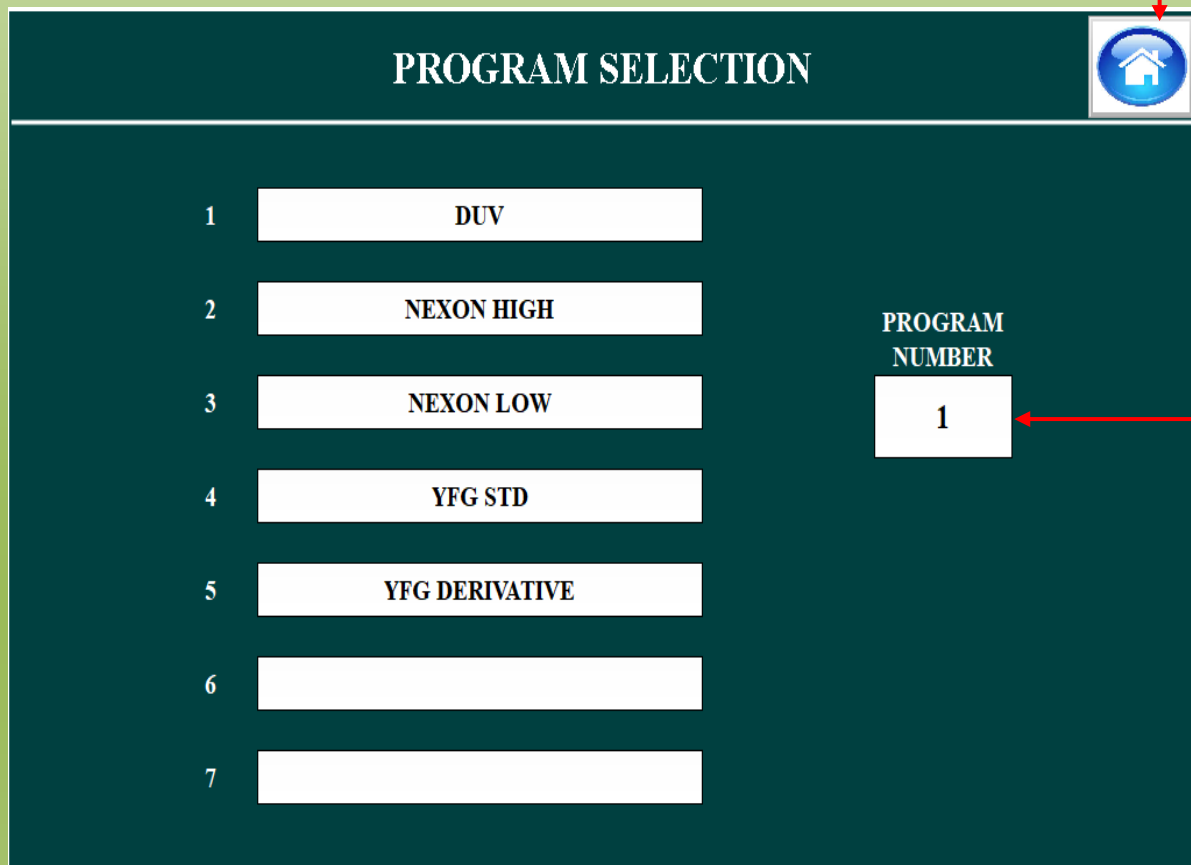
4. Scara robot parameter manual operation buttons.

5. Index home status & start/stop manual operation button.

6. Buzzer on/off manual button.

7. Screwdriver status & enable/disable manual operation button.
8. LH and RH camera ok, no status, scene number setting, trigger manual operation button.
9. Visco gluing parameters status and manual operation buttons.
10. Pallate-A Clamp cylinder and manual flap reed switch status, clamp cylinder fwd/ret manual operation button.
11. Pallate-A LH and RH cavity sensors status.

PROGRAM SELECTION SCREEN



PROGRAM SELECTION

1	DUV	PROGRAM NUMBER 1
2	NEXON HIGH	
3	NEXON LOW	
4	YFG STD	
5	YFG DERIVATIVE	
6		
7		

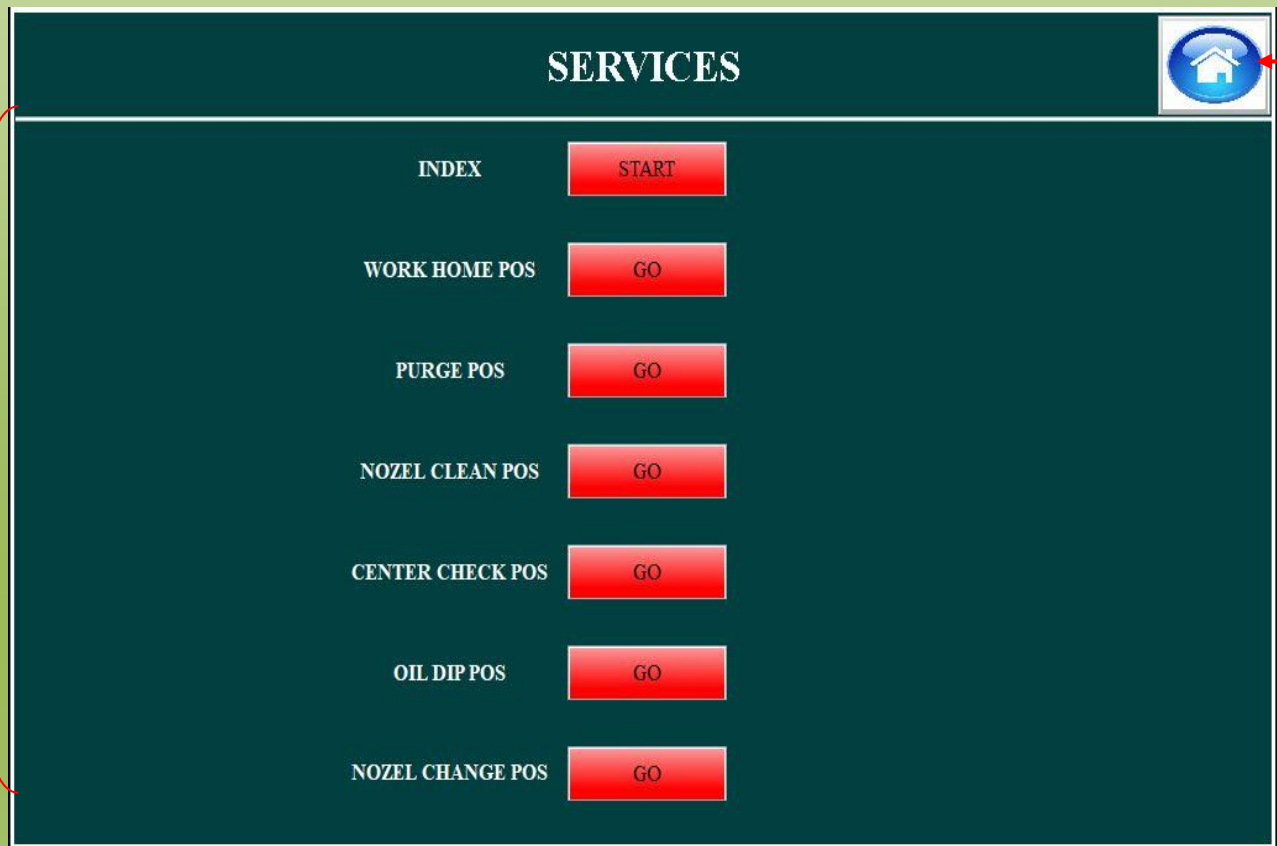
1. Home screen navigation button.
2. Program number setting.
3. Program selection list.

ROBOT PROGRAM NUMBER SCREEN

ROBO_PROG_NO					
LH CAVITY		RH CAVITY			
GLUE POS-1		PROG NO. 7	GLUE POS-1		PROG NO. 6
GLUE POS-2		0	GLUE POS-2		0

1. Home screen navigation button.
2. RH cavity glue position-1, 2 select/unselect & program number selection.
3. LH cavity glue position-1, 2 select/unselect & program number selection.

SCARA ROBOT SERVICES SCREEN



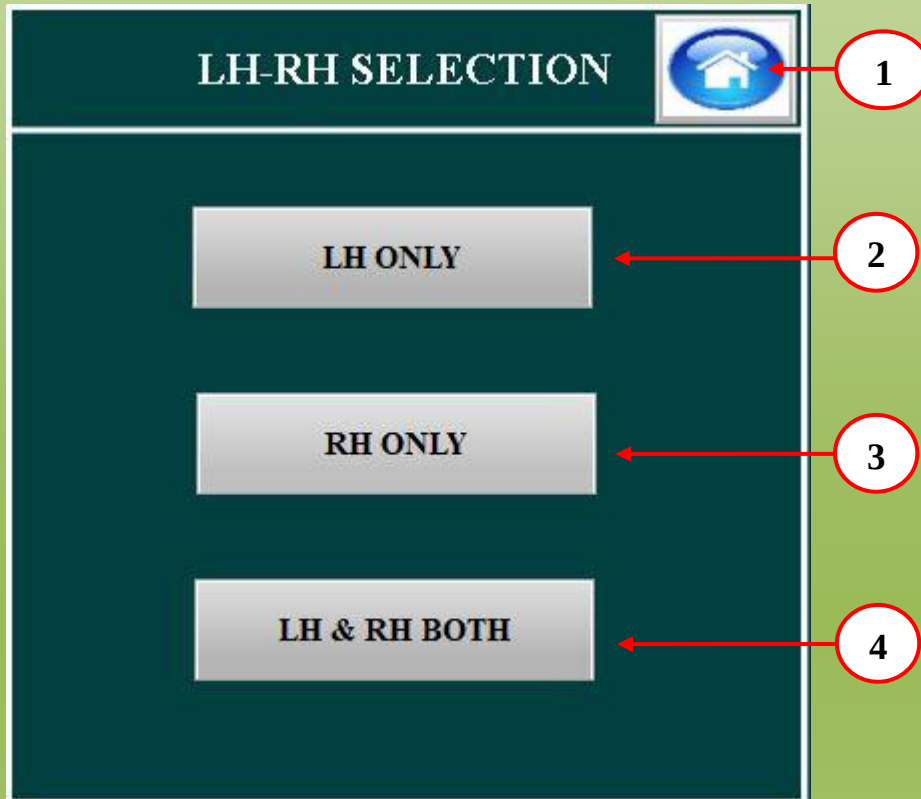
1. Home screen navigation button.
2. Scara robot related services operation buttons.

NOZZEL CHANGE & WEIGHT CHECK SCREEN

The screenshot shows a control interface for a thermal glue dispensing machine. It is divided into two main sections: 'NOZZEL CHANGE' on the left and 'WEIGHT CHECK' on the right. At the top right is a home icon (1). The 'NOZZEL CHANGE' section includes buttons for 'NOZZEL CHANGE', 'OLD NOZZEL REMOVED', 'NEW NOZZEL INSERTED', and 'PURGE COMPLETE', each with a 'START' button. There are also 'NOZZEL SENSOR' indicators (8) and a 'NO ERROR' status bar (6). The 'WEIGHT CHECK' section includes 'ROBO HOME' and 'INDEX' status indicators with 'GO' buttons (2), a 'WEIGHT CHECK' status indicator with a 'START' button (3), and displays for 'SET WEIGHT', 'ACT GLUE WEIGHT', and 'GLUE WEIGHT' (4). It also shows 'SHIFT A WEIGHT', 'SHIFT B WEIGHT', and 'SHIFT C WEIGHT' (5). A 'NO ERROR' status bar is at the bottom (6). A red arrow (7) points to the 'AUTO GLUING' button. Numbered callouts 1 through 9 are placed around the screen to identify specific elements.

1. Home screen navigation button.
2. Robot home, index home status & manual operation buttons.
3. Weight checks status & start/stop operation button.
4. Set weight, actual weight & glue weight values display in milligram.
5. Shift A, B & C weight display in milligram.
6. Error message display.
7. Auto gluing start/stop operation button.
8. Nozzle sensor status.
9. Nozzle parameter status & operation buttons.

LH-RH SELECTION SCREEN



1. Home screen navigation button.
2. LH side fixture selection button.
3. RH side fixture selection button.
4. LH & RH side fixture selection button.

INPUT STATUS SCREEN

INPUT STATUS			
			
 X0: START PB-1	 X30: VISCO-INPUT PRESSURE OK	 X50: FIX-B RH PP SENS-1	 X70: CAM-1 OK
 X1: START PB-2	 X31: VISCO-CRTRIDGE NOT EMPTY	 X51: FIX-B RH PP SENS-2	 X71: CAM-1 RUN/BUSY
 X2: EMG STOP PB	 X32: VISCO-READY	 X52: FIX-B FLAP FWD	 X72: CAM-1 NOK
 X3: RESET PB	 X33: VISCO-HEALTHY	 X53: FIX-B FLAP RET	 X73: CAM-2 OK
 X4: MACHINE DIGITAL PSW	 X34: VISCO-SPARE	 X54: FIX-B LOCKING CYL FWD RD	 X74: CAM-2 RUN/BUSY
 X5: VISCO DIGITAL PSW	 X35: INDEXER HOME SENS	 X55: FIX-B LOCKING CYL RET RD	 X75: CAM-2 NOK
 X6: LIGHT CURTAIN FB	 X36: FIX-A LH PP SENS-1	 X56: HOMMING SENSOR-1	 X76: PALLET-A FIX FROM INDO
 X7: REJECTION BIN SENS	 X37: FIX-A LH PP SENS-2	 X57: HOMMING SENSOR-2	 X77: PALLET-B FIX FROM INDO
 X20: EPSON READY	 X40: FIX-A RH PP SENS-1	 X60: DOOR LOCK LIMIT SW	 X100: FIX-A FIX SEL-1
 X21: EPSON RUNNING	 X41: FIX-A RH PP SENS-2	 X61: DOOR LOCK LIMIT SW	 X101: FIX-A FIX SEL-2
 X22: EPSON PAUSE	 X42: FIX-A FLAP FWD	 X62: SCREWDRIVER OK	 X102: FIX-A FIX SEL-4
 X23: EPSON ERROR	 X43: FIX-A FLAP RET	 X63: SCREWDRIVER NOK	 X103: FIX-A FIX SEL-8
 X24: EPSON E-STOP ON	 X44: FIX-A LOCKING CYL FWD RD	 X64: VFD ERROR	 X104: FIX-B FIX SEL-1
 X25: EPSON POS REACHED	 X45: FIX-A LOCKING CYL RET RD	 X65: VFD RUNNING	 X105: FIX-B FIX SEL-2
 X26: EPSON DISPENSING ON	 X46: FIX-B LH PP SENS-1	 X66: FIX-B AT OPERATOR POS	 X106: FIX-B FIX SEL-4
 X27: EPSON AT HOME POS	 X47: FIX-B LH PP SENS-2	 X67: FIX-A AT OPERATOR POS	 X107: FIX-B FIX SEL-8

1. Home screen navigation option.
2. Plc input status.

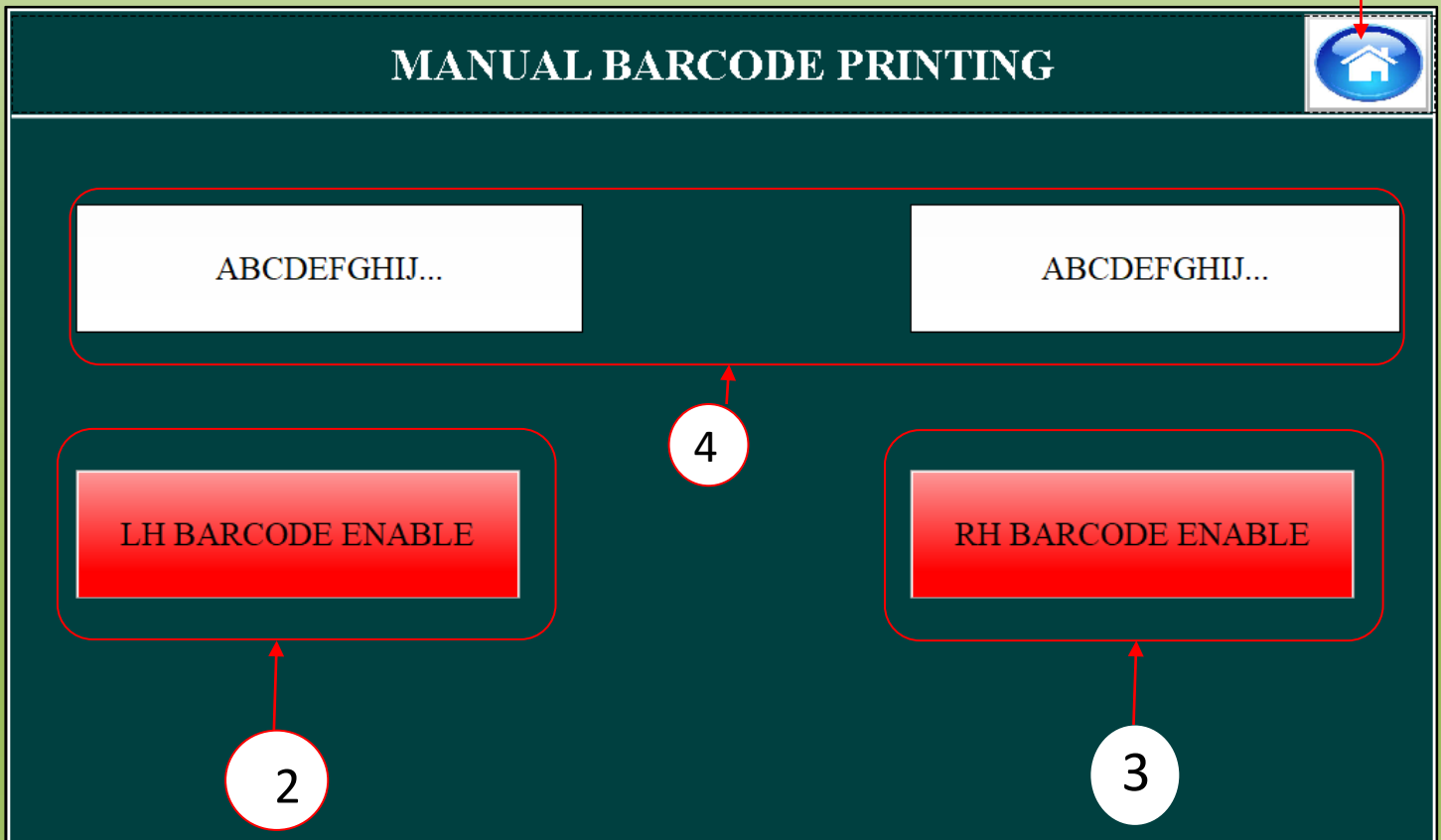
OUTPUT STATUS SCREEN

OUTPUT STATUS			
			
 Y0: PASS LAMP	 Y30: EPSON SPARE	 Y50: VFD START CCW	 Y70: LH CAM SCENE SEL-1
 Y1: ERROR LAMP	 Y31: EPSON SPARE	 Y51: VFD SPEED SELECT	 Y71: LH CAM SCENE SEL-2
 Y2: PROCESS LAMP	 Y32: VISCO- START DISPENSING	 Y52: VFD BREAK	 Y72: LH CAM SCENE SEL-4
 Y3: BUZZER	 Y33: VISCO- ERROR RESET	 Y53: EPSON RESET	 Y73: LH CAM CMD LOW
 Y4: NA	 Y34: VISCO INPUT-3	 Y54: EPSON PROG SEL-16	 Y74: LH CAM CMD HIGH
 Y5: NA	 Y35: FIX-A LOCKING CYL FWD SOL	 Y55: EPSON FIX-A INDICATION	 Y75: LH CAM EXECUTE
 Y6: NA	 Y36: FIX-A LOCKING CYL RET SOL	 Y56: EPSON MOV TO NEXT POS	 Y76: SPARE
 Y7: NA	 Y37: FIX-B LOCKING CYL FWD SOL	 Y57: SICK SCANNER TRIGGER	 Y77: SPARE
 Y20: EPSON START	 Y40: FIX-B LOCKING CYL RET SOL	 Y60: RH CAM TRIGGER	 Y100: NA
 Y21: EPSON PROG SEL-1	 Y41: SCREWDRIVER ENABLE	 Y61: RH CAM SCENE SEL-1	 Y101: NA
 Y22: EPSON PROG SEL-2	 Y42: SPARE	 Y62: RH CAM SCENE SEL-2	 Y102: NA
 Y23: EPSON PROG SEL-4	 Y43: MAIN AIR OFF	 Y63: RH CAM SCENE SEL-4	 Y103: NA
 Y24: EPSON PROG SEL-8	 Y44: VISCO AIR OFF	 Y64: RH CAM CMD LOW	 Y104: NA
 Y25: EPSON STOP	 Y45: PANEL FAN-1	 Y65: RH CAM CMD HIGH	 Y105: NA
 Y26: EPSON PAUSE	 Y46: PANEL FAN-2	 Y66: CAM-1 EXECUTE	 Y106: NA
 Y27: EPSON CONTINUE	 Y47: VFD START CW	 Y67: LH CAM TRIGGER	 Y107: NA

1. Home screen navigation option.

2. Plc Output status.

MANUAL BARCODE PRINTING SCREEN



1. Home screen navigation option.
2. LH Barcode Print Enable button.
3. RH Barcode Print Enable Button.
4. Barcode number.

AM PM CHECK LIST:

YFG THERMAL GLUE DISPENSING MACHINE AM PM CHECK POINTS							
General points	Points to be check	Where to check	Daily	Monthly	Quarterly	Biannual	Annual
1	Check given power supply and earthing .ensure it is within range as per machine required	Control panel	√	√	√	√	√
2	Check Pnumatic pressure .ensure it is within range as per required	Pnumatic device panel	√	√	√	√	√
3	Check looseness of electric connectors. If the connectors are loosen, push it securely or tighten	External connector terminal mounted on back side of monitor, Electric indo connector		√	√	√	√
4	Check looseness of punmatic connector, If the connectors are loosen, push it securely or tighten	On machine side & fixture side punmatic connector		√	√	√	√
5	Check computer cpu on and shut off as per given process. Cpu Improper on /off will cause system crash. CPU on process 1.On power supply of machine . 2.After that power on ups . 3.Then on CPU. 4.After windows boot properly open application as required. CPU shut down process 1.Close all application properly. 2.Then go to windows menu and tab shut down button. 3.After windows properly off shut off ups power	Computer unit	√	√	√	√	√

6	Check printer are properly calibrate ,refer printer manual for calibration	Printer	√	√	√	√	√
7	Check printer label sensor .			√	√	√	√
8	Check printer power cable & hmi printer communication cable.			√	√	√	√
9	Check scanner communication cable	Scanner		√	√	√	√
10	Check barcode scanning properly, if it is any issue to scan calibrate it. Refer scanner manual.		√	√	√	√	√
11	Check communication cable of camera	Camera to PLC		√	√	√	√
12	Check looseness of camera mounting bolts, screws. Tighten them if necessary as per match marking.					√	√
13	Check camera teaching using master parts if it is not working, teach camera under supplier guidance.			√			
14	Check looseness or backlash of bolts, screws. Tighten them if necessary. Use robot maintenance manual.	End effectors mounting bolts ,Manipulator mounting bolts , Each arm locking bolts ,Bolts, screws around shaft ,Bolts, screws securing motors, reduction gear units, et		√	√	√	√
15	Check looseness of connectors. If the connectors are loosen, push it securely or tighten	External connectors on Manipulator (on the connector plates etc.) Manipulator cable unit		√	√	√	√
16	Check robot teaching points using master part, if it is not working ,teach robot under supplier guidance,			√	√	√	√



YFG HL THERMAL GLUE DISPENSING MACHINE

MARELLI MOTHERSON AUTO SYSTEM LTD.
GAT NR. 148-150, VILLAGE AMBETHAN TALUKA KHED,
NEAR DWARKA SCHOOL, PUNE, MAHARASHTRA- 410501



17	Wipe the surface with a soft, dry cloth		√	√	√	√	√
18	Check glue is not to be dry, if it is dry clean all viscotec mechanism .refer viscotec maintenance manual		√	√	√	√	√
19	Check glue calibration		√	√	√	√	√
20	Check glue dispenser air pressure it should be under 2bar		√	√	√	√	√
21	Oil change is required after 3 years of usage of cam index. Use recommended oil refer Maintenance manual of cam index	Cam index drive					√

DO'S & DONT'S:

Every effort has been taken to ensure user safety while designing this machine. However, you are requested to adhere to following instructions to avoid any mishap.

- The machine should not be used for any other purpose other than that specified in the intended use.
- Do not operate the machine without understanding the installation, operation and maintenance manual thoroughly.
- Only trained, skilled, qualified and authorized personnel should install, operate and maintain the machine.
- Do not commission the machine without the aid of the qualified instructor.
- Do not reach into moving machinery.
- Never perform complex maintenance procedure single handed. Especially during start-ups as it is necessary to have another person manning the emergency stop in case of any problem.
- No tools other than that are prescribed should be used for machine maintenance.
- Do not store any loose tools in the machine compartments or control panel.
- If the machine is down for maintenance, place a sign over the control panel to inform others about that.
- Switch off the machine while connecting/disconnecting into electrical connector.



YFG HL THERMAL GLUE DISPENSING MACHINE

MARELLI MOTHERSON AUTO SYSTEM LTD.
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